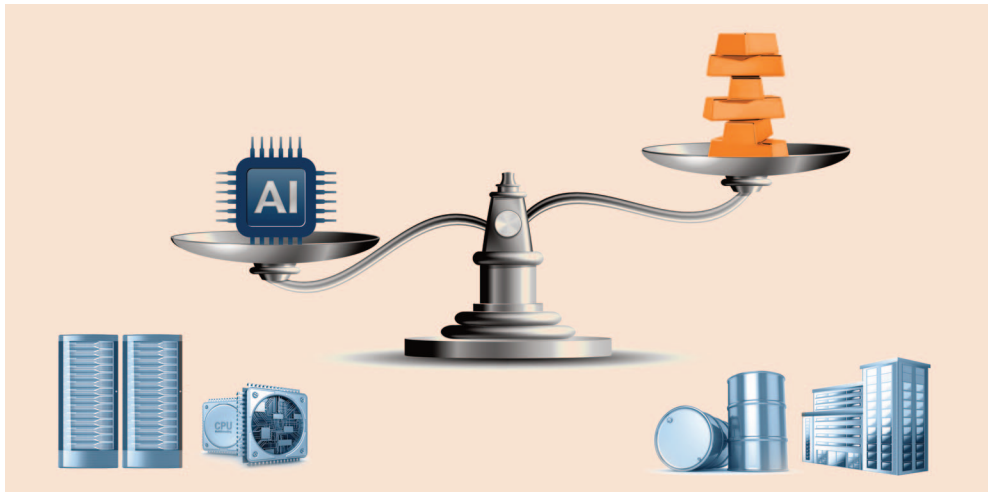


GLOBAL STRATEGY PAPER NO. 77

Balancing Innovation and Inflation in Portfolios



- Global financial assets have performed strongly in the last 3 years after the large drawdown in balanced portfolios in 2022. Equities have performed particularly strongly, with long-term ex post equity risk premia nearing the Golden 20s and 50s and the AI boom boosting Tech stocks. As a result, the 'World Portfolio' has drifted towards US assets, equities and the Tech sector in recent years.
- Structural cycles linked to innovation or inflation have historically often driven prolonged trends in performance across assets, regions and sectors and, as a result, the composition of the 'World Portfolio' has materially drifted – increasing risk in the case of structural breaks. The current AI capex boom increases the risk that falling profitability for mega-cap Tech stocks materially drags on equity returns before benefits from AI adoption show up.
- And higher inflation volatility and fiscal risks are also creating headwinds for balanced portfolios, with less of a buffer from bonds and more risk of rate shocks. Based on our macro-based return forecasting framework, in all but an optimistic 'Goldilocks + AI boom' scenario, expected long-term equity returns are below the long-run average, pointing to lower equity allocations.
- The composition of market value-weighted global multi-asset benchmarks drifts due to asset supply and relative performance and, as a result, it is often backward-looking. Regularly rebalanced 'regime-neutral' portfolios outperformed the 'World Portfolio' over the long run and currently point to lower weights in equities and Tech and higher allocations to real assets.
- However, during strong equity bull markets, especially if they are driven by the Tech sector, it was difficult to lean against momentum. In the short term, being underinvested can be costly, with equities usually performing well up until the bull market peak. Investors need to keep exposure to innovation, protect against inflation and improve risk mitigation – we recommend five strategies to stay invested while balancing innovation and inflation in global multi-asset portfolios.

Christian Mueller-Glissmann, CFA
+44(20)7774-1714
christian.mueller-glissmann@gs.com
Goldman Sachs International

Andrea Ferrario
+44(20)7552-4353
andrea.ferrario@gs.com
Goldman Sachs International

Alessandro Giglio
+44(20)7051-6240
alessandro.giglio@gs.com
Goldman Sachs International

Giovanni Ferrannini
+44(20)7051-2589
giovanni.ferrannini@gs.com
Goldman Sachs International

Elena Porfidia
+44(20)7051-5240
elena.porfidia@gs.com
Goldman Sachs International

Peter Oppenheimer
+44(20)7552-5782
peter.oppenheimer@gs.com
Goldman Sachs International



For our asset allocation frameworks, see our dedicated [Balanced Bear page >](#)

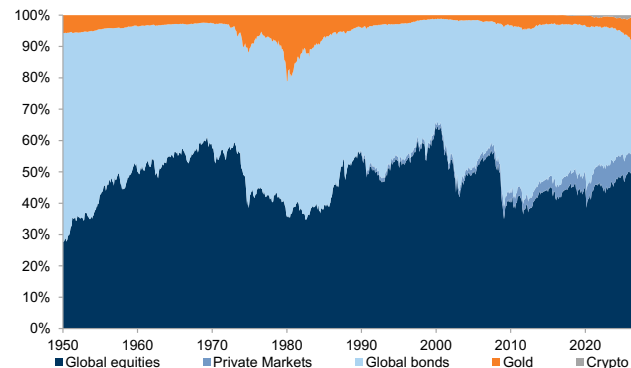
This report is intended for distribution to GS institutional clients only.

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

Summary

- Global financial assets have performed strongly in the last 3 years after the large drawdown in balanced portfolios in 2022. Our 'World Portfolio' or a US 60/40 portfolio delivered unusually strong real returns since the GFC, comparable to the Golden 20s and 50s.
- Ex post equity risk premia have also been particularly high and were boosted by strong growth in the Tech sector, in particular in the US and, more recently, in Asia. As a result, the composition of the 'World Portfolio' has drifted towards US assets (also supported by a stronger Dollar since the GFC), equities and the Tech sector.
- Innovation and inflation have been among the most important drivers of structural cycles. Those structural cycles have historically driven prolonged trends in performance across assets, regions and sectors and, as a result, major drifts in the composition of the 'World Portfolio'. During periods of innovation, equities tend to perform well, particularly growth stocks and the Tech sector, and balanced portfolios deliver the strongest Sharpe ratios. However, balanced portfolios often eventually become imbalanced.
- The current AI capex boom increases the risk that falling profitability for mega-cap Tech stocks materially drags on equity returns before benefits from AI adoption show up. Higher inflation volatility and fiscal risks also create headwinds for balanced portfolios with less of a buffer from bonds and risk of rate shocks. Based on our long-term return forecasting framework, in all but a very bullish 'Goldilocks + AI boom' scenario, expected long-term equity returns are below the long-run average.
- The composition of market value-weighted global multi-asset portfolios drifts due to asset supply and relative performance and, as a result, is often backward-looking. Optimal portfolios, in contrast, should be conditioned on future macro regimes.
- Regularly rebalanced 'regime-neutral' portfolios outperformed the 'World Portfolio' over the long run and currently point to lower weights in equities and Tech and higher allocations to real assets. However, during strong equity bull markets, especially if they are driven by the Tech sector, it was difficult to lean against momentum.
- However, even though valuations are high and uncertainty around both innovation and inflation remains high, it remains difficult to lean against the current momentum – time in the market competes with timing the market. We find that equities deliver some of their strongest returns in the final years of a bull market, often led by the sector that outperformed in the preceding years.
- Investors need to keep exposure to innovation, protect against inflation and improve risk mitigation in portfolios. We recommend five strategies that help investors stay invested and achieve those objectives: (1) selective real asset allocations, (2) opportunities in factors and style diversification, (3) regional diversification opportunities across assets, (4) option strategies such as long-dated calls and (5) selective alternatives that are uncorrelated with the 'World Portfolio'.

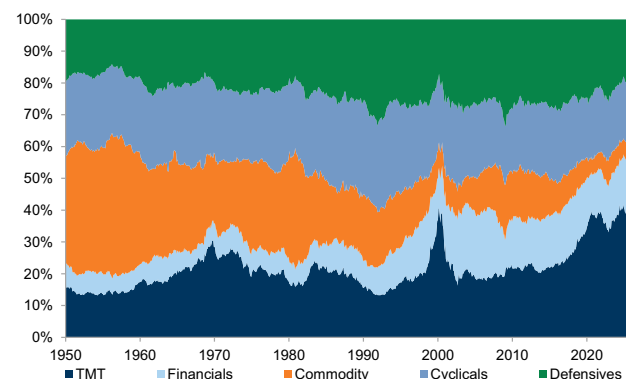
Exhibit 1: The World Portfolio is shifting further to equities, especially when including private markets



Source: Bloomberg, Datastream, Haver Analytics, Preqin, Goldman Sachs Global Investment Research

Exhibit 2: The Tech sector equity weight is now above Tech Bubble levels

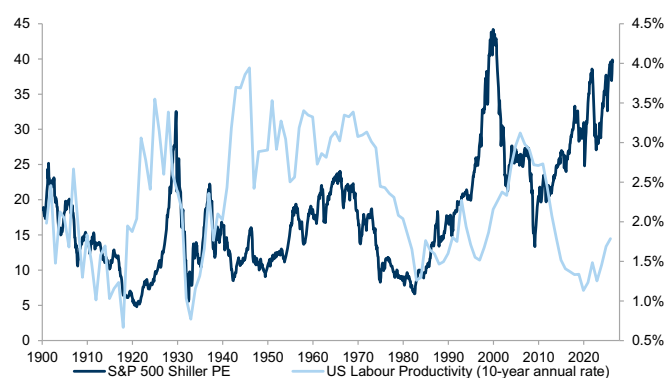
Relative sector weights within US equities



Source: Bloomberg, Datastream, Kenneth French, Goldman Sachs Global Investment Research

Exhibit 3: S&P 500 Shiller P/E's are nearing the Tech Bubble levels

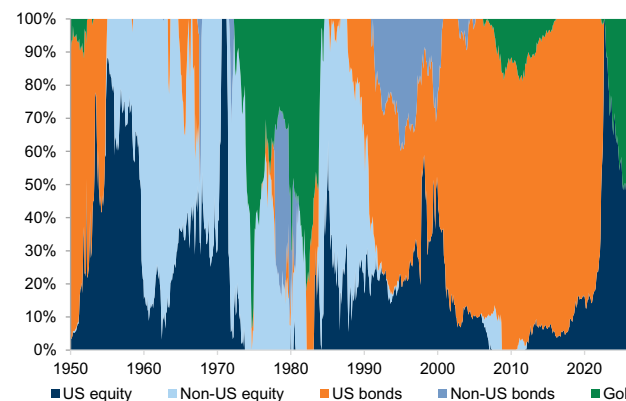
S&P 500 Shiller P/E vs. US labour productivity growth



Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: The optimal portfolio shifted notably over time, deviating materially from the World Portfolio benchmark

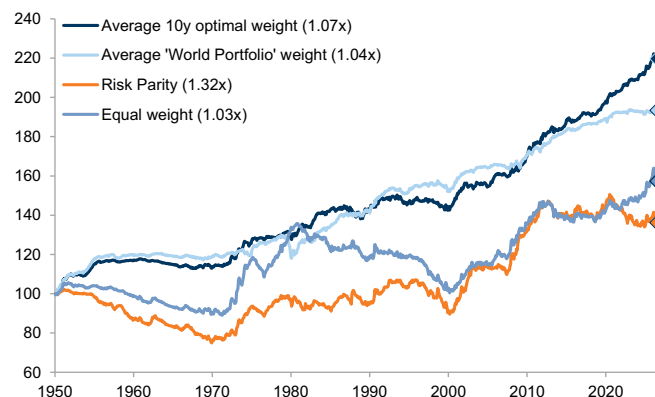
10-year rolling weight of highest Sharpe ratio portfolio



Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Exhibit 5: Regime-neutral portfolios outperformed on a risk-adjusted basis in the medium term

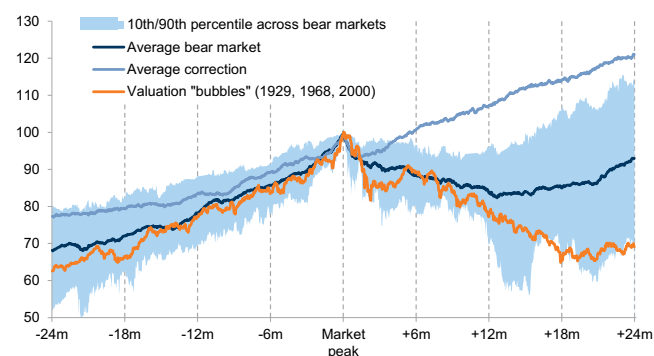
Relative return vs World Portfolio (risk-neutral scaling to match the World Portfolio vol, in brackets)



Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: Equities tend to perform well into bull market peaks and corrections tend to be short-lived

Average S&P 500 total return performance around bear markets and corrections



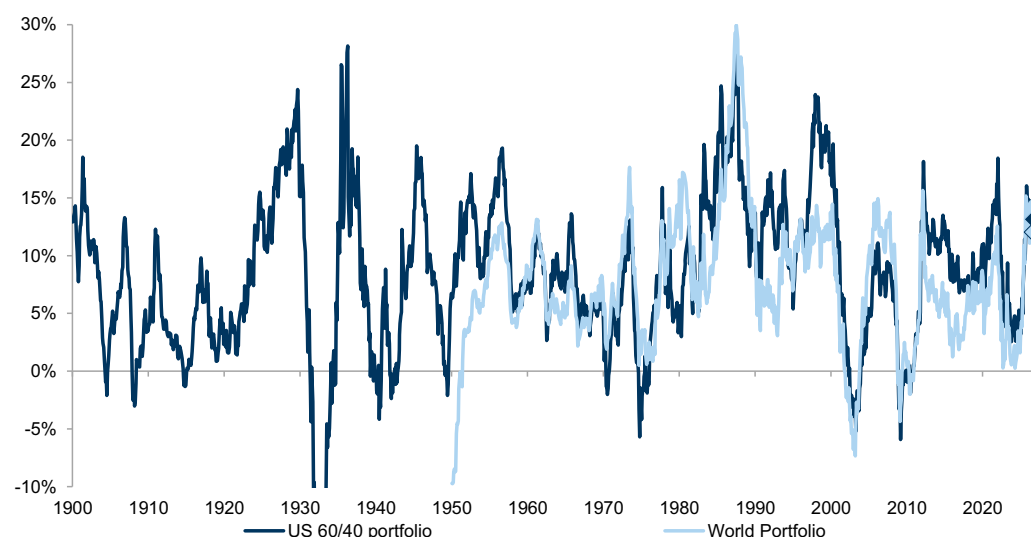
Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Imbalanced portfolios — Tech stock momentum becomes key driver

Bull market in financial assets has resumed

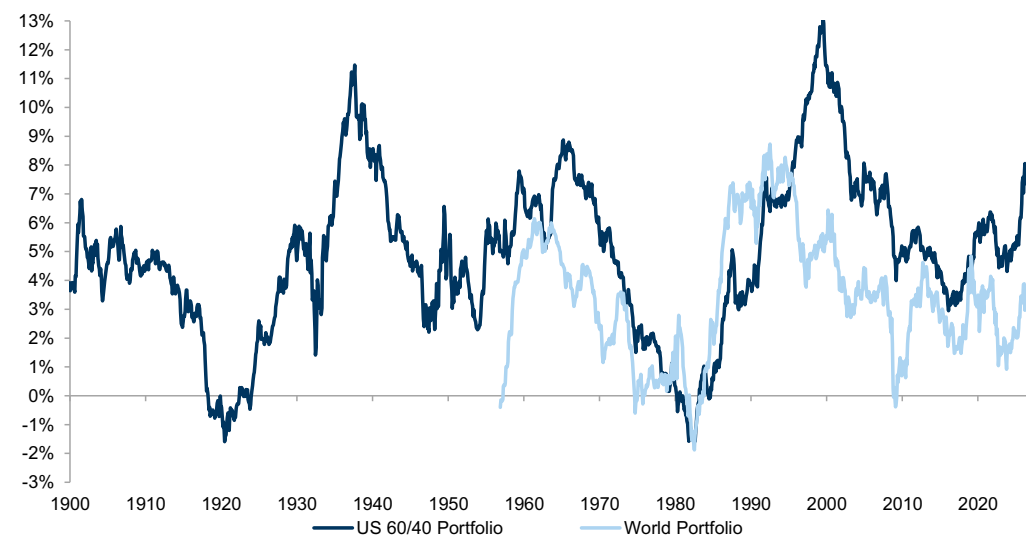
In the last 3 years, financial assets recovered strongly after the large drawdown in balanced portfolios ('Balanced Bear') in 2022. In [Global Strategy Paper: Investing in Everything, Everywhere, All at Once](#), we introduced a 'World Portfolio' benchmark tracking roughly US\$300tn of global financial assets. Over the last 3 years, the returns of this 'World Portfolio' have been among the strongest since WW2 ([Exhibit 7](#)). Similarly, a US 60/40 portfolio, which we can track back to 1900, has seldom delivered stronger returns outside of the Golden 20s, 50s, the early 1980s and the late 1990s.

Exhibit 7: Balanced portfolios have delivered strong returns in recent years
3-year rolling annualised nominal return (US\$)



Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 8: One of the strongest bull markets in financial assets post GFC, led by the US
10-year rolling real return (US\$)

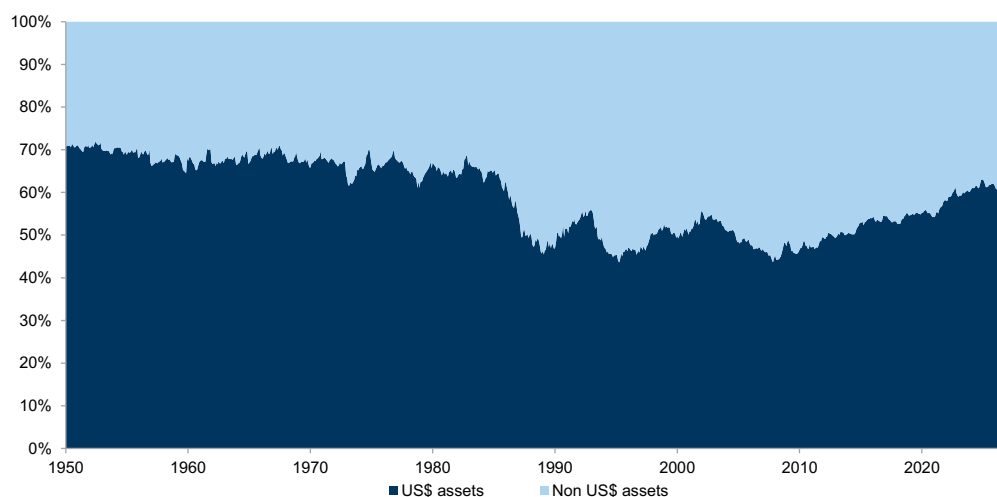


Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Balanced portfolios, especially in the US, are closing in on the longest and strongest periods of long-run performance on record – the last decade’s real returns for a US 60/40 portfolio are close to 8%, or nearly twice the long-run average. There have only been similarly strong periods of real returns during the Golden 20s and 50s, and the 1990s. On the flipside, there were several periods of unusually poor real returns, such as around WW1 and WW2, as well as the 1970s and the 2000s Tech Bubble Burst. The ‘World Portfolio’ performed less strongly in real terms, as non-US assets generally underperformed US assets, in part due to a strong Dollar.

Our ‘World Portfolio’ benchmark is now roughly half in US\$ assets and half in non-US\$ assets. US\$ assets have been a dominant driver of growth in financial assets post GFC – their proportion has steadily increased after declining during the 2000s ([Exhibit 9](#)). Post GFC, the US\$ bond weight picked up owing to higher yields relative to the Rest of the World, given a better post-GFC recovery, the Fed’s avoidance of negative rates, and a strong Dollar. Since the GFC, US equities have materially outperformed non-US markets, at least until recently – as a result, the MSCI AC World equity benchmark currently has a c.63% US weight.

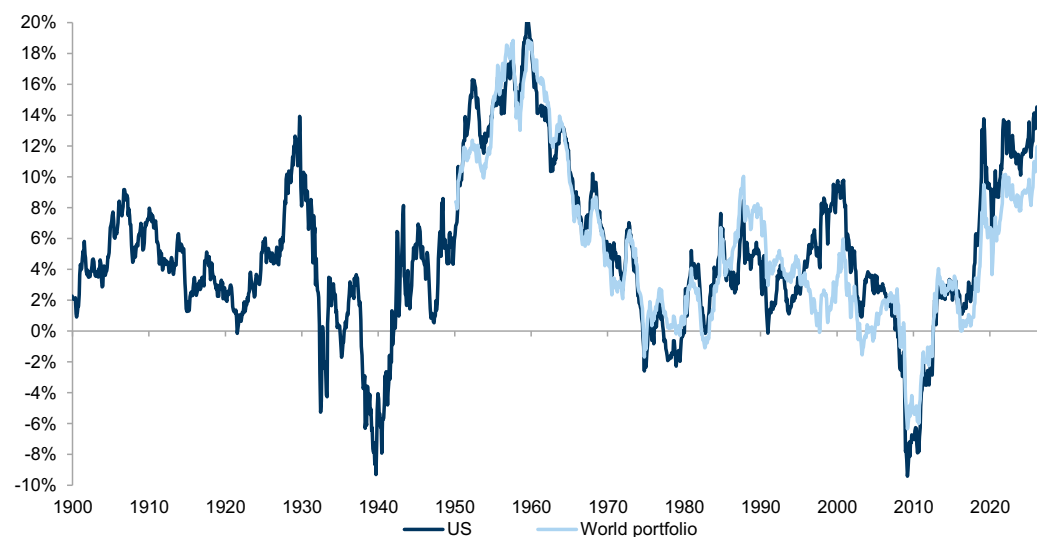
Exhibit 9: US assets have become more dominant in the World Portfolio again
Relative asset weights in the World Portfolio



Source: Bloomberg, Datastream, Preqin, World Gold Council, Goldman Sachs Global Investment Research

High equity risk premia boost equity weight in the ‘World Portfolio’
Ex post equity risk premia have been very high and were boosted further by poor bond returns since the COVID-19 crisis ([Exhibit 10](#)). The last decade’s ex post equity risk premia are well above the long-run average of 4.5–5%. In fact, the annualised US equity vs. bond performance has been nearly three times the long-term average. The only prolonged periods of similarly high equity risk premia were during the Golden 20s and 50s. Equities also performed well during both the early 1980s and the late 1990s, but the outperformance vs. bonds was less pronounced. Poor bond returns in recent years, due to very low starting yields after the COVID-19 crisis and higher inflation and fiscal risks since then, have added to ex post equity risk premia.

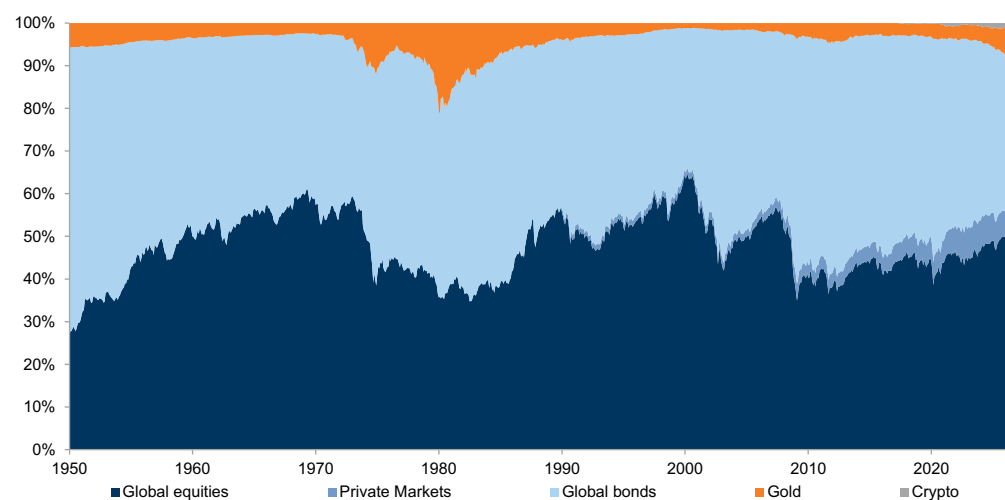
Exhibit 10: Equity risk premia have been among the strongest since the Golden 20s and 50s
10-year rolling annualised equity vs. bond performance (ex post equity risk premia)



Source: Robert Shiller, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

As a result, the equity weight in the ‘World Portfolio’ is nearing Tech Bubble peaks, especially when private markets are included (Exhibit 11). In the 2000s, the equity weight trended down when the Tech Bubble burst and again following the GFC. However, since the GFC, strong performance has boosted the equity weight once again. Although equities have since regained some ground, their share remains below late-1990s levels, which were further boosted by the IPO boom of that period; in recent years, buybacks have dominated instead. However, when including private markets, which contain several large Tech unicorns that might IPO in the coming years, the ‘World Portfolio’ weight in equities is closing in on the Tech Bubble peak.

Exhibit 11: The World Portfolio is shifting further to equities, especially when including private markets



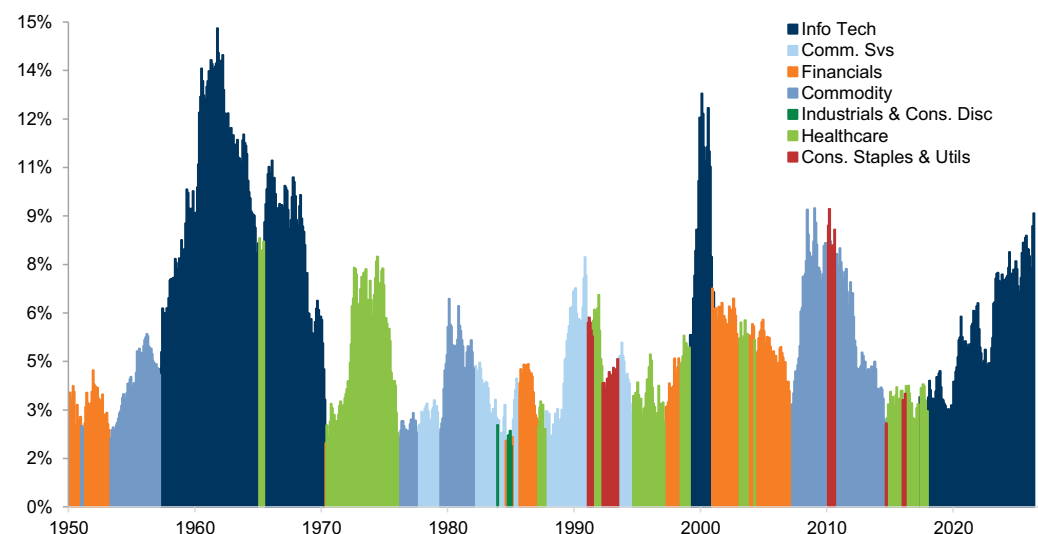
Source: Bloomberg, Datastream, Haver Analytics, Preqin, Goldman Sachs Global Investment Research

Technology dominance has driven equity benchmark concentration

Strong equity performance has been boosted by the Technology sector, which has become increasingly dominant within equities (Exhibit 12). Structural cycles linked to the Technology sector have historically often been a key driver of the equity weight, notably during the PC boom of the 1960s, the 1990s Internet bubble, and the rise of the Magnificent 7, or 'hyperscalers', since the GFC. Different sectors have also regularly dominated equity performance historically – Commodities sectors were a dominant driver during the 1950s, the 1970s and around the GFC, while Financials outperformed in the 2000s, boosted by increased financialisation of the global economy.

Exhibit 12: Tech sector outperformance has been unusually strong and prolonged

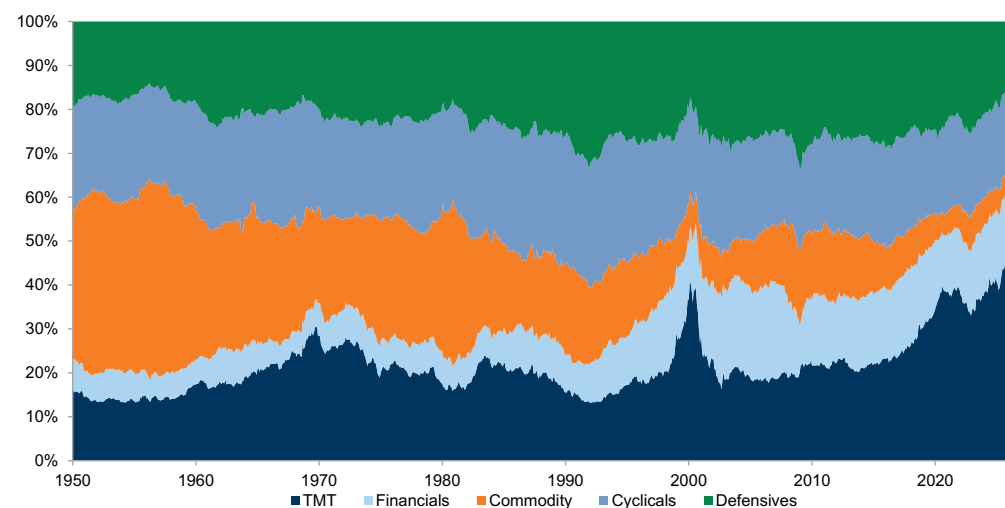
10-year annualised performance vs. the market of the best performing US sector



Source: Kenneth French, Goldman Sachs Global Investment Research

Exhibit 13: The Tech sector equity weight is now above Tech Bubble levels

Relative sector weights within US equities



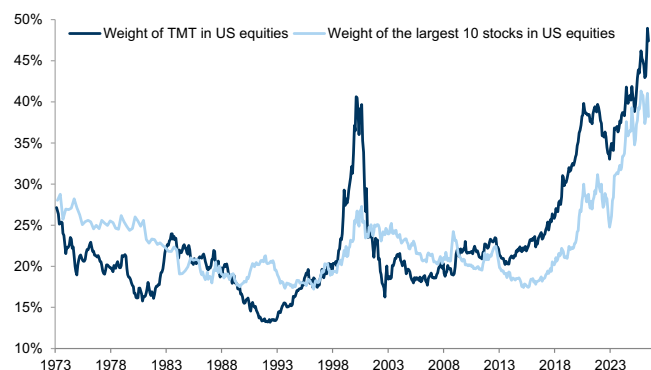
Source: Bloomberg, Datastream, Kenneth French, Goldman Sachs Global Investment Research

Since the GFC, the global TMT sector (combining Technology and Communication Services) has gained material share in the global equity benchmark. More recently optimism around AI has supported Tech sector returns and there have been more

capital raises and IPOs – the current weight in the US equity benchmark is already above the Tech Bubble peak ([Exhibit 13](#)). However, in contrast to the Tech Bubble, the Tech sector weight has increased over multiple years, indicating that it has been more closely linked to fundamentals. Financials, which lost ground after the GFC, have also started to recover since the COVID-19 crisis, while Commodity and Defensive sectors such as Healthcare and Utilities have lost share.

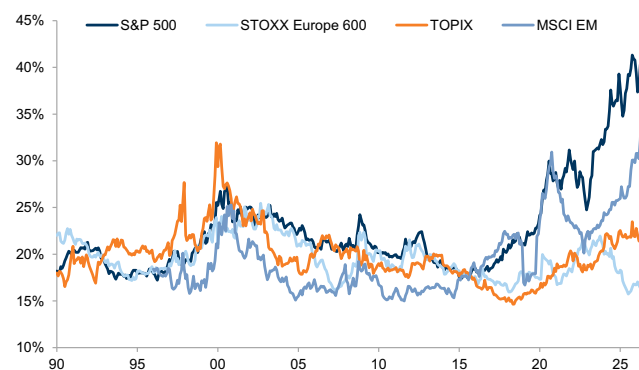
A consequence of the growing Technology sector has been a higher concentration in a few large cap stocks, which have contributed significantly to strong index returns ([Exhibit 14](#)). The same was true during the Tech Bubble, but earnings are now much more concentrated. Still, depending on the degree of correlation among the constituent companies and with the index itself, greater index concentration may reduce diversification benefits and increase overall risk. More recently, Tech dominance has also increased in Asia, with narrow performance led by AI capex beneficiaries, especially semiconductors ([Exhibit 15](#)). While the ‘hyperscalers’ in the US have historically been less correlated and less cyclical, with strong balance sheets and cash flows generation¹, semiconductor companies in North Asia are highly correlated and cyclical businesses.

Exhibit 14: Concentration in the US equity indices has increased alongside the Tech sector weight
Weight in US equity benchmark index



Source: Datastream, Goldman Sachs Global Investment Research

Exhibit 15: Stock concentration has increased both in the US and, more recently, in EM benchmarks
Weight in the largest 10 stocks



Source: Datastream, Goldman Sachs Global Investment Research

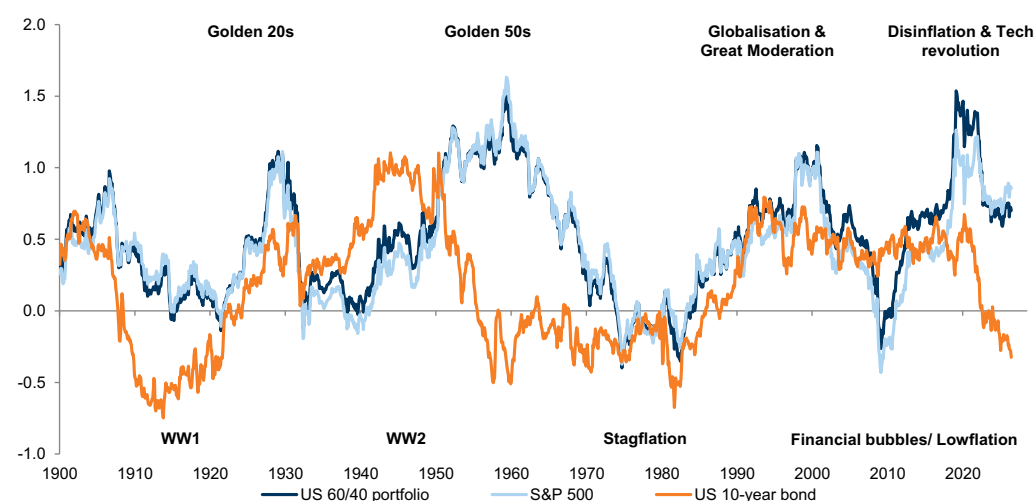
¹ In fact Kritzman and Turkington (2026) argue that the US stock market has not become riskier as it has become more concentrated, in part as larger companies are more diversified and resilient – they show that since 1926 their increased weights are offset by their lower volatility compared to small stocks.

Innovation vs. Inflation as structural drivers of balanced portfolios

Structural cycles and balanced portfolio performance

Innovation and inflation have among been the most important drivers of trends across asset classes, regions and sectors. Slow-moving trends in growth, inflation and policy can result in structural cycles – these are often linked to demographics and social attitudes, technological developments and productivity trends, as well as policy choices and regimes. 60/40 portfolios delivered the highest returns during periods of successful innovation with anchored inflation and rising productivity growth, such as the Golden 20s and 50s, the 1990s and the post-GFC period. On the flip side, prolonged periods of high inflation, often due to wars, resulted in ‘lost decades’ ([Exhibit 16](#)).

Exhibit 16: US 60/40 portfolio performance varied materially owing to structural cycles
10-year rolling Sharpe ratio



Source: Haver Analytics, Robert Shiller, Goldman Sachs Global Investment Research

Balanced portfolios perform best in a structural ‘Goldilocks’ scenario with low inflation and good growth – as [Exhibit 17](#) shows, the sweet spot occurs when the economy is neither too hot nor too cold. With low and anchored inflation, bonds tend to act as a buffer for equities during growth shocks, resulting in better risk-adjusted returns for US 60/40 portfolios. On average, returns declined with lower trend growth and real returns were poor during stagflationary episodes. A favourable structural growth/inflation mix also tends to boost equity risk premia.

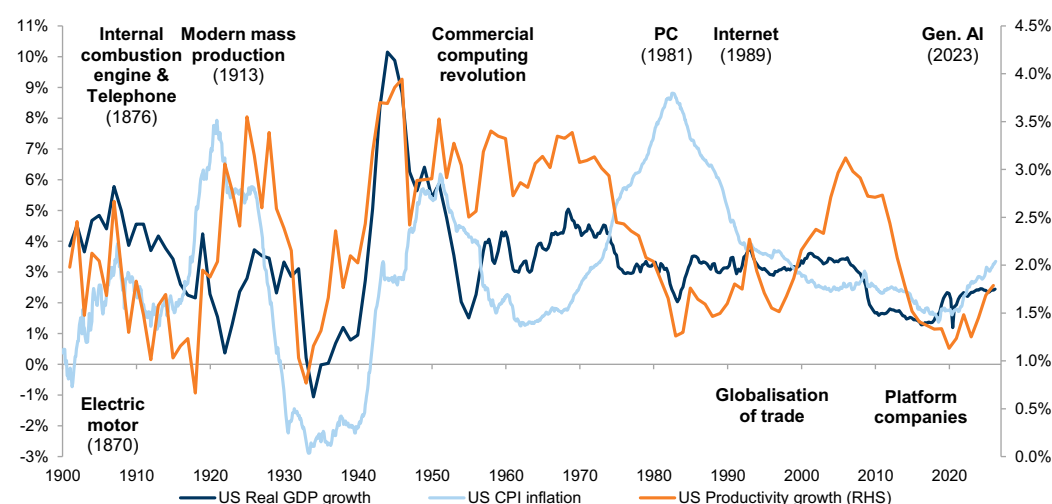
Exhibit 17: Balanced portfolios performed best in a structural ‘Goldilocks’ scenario with low inflation and good growth
10-year rolling performance in function of 10-year rolling real GDP growth/ headline CPI inflation

US 60/40 real return					US 60/40 Sharpe ratio					Realised Equity Risk Premium							
Inflation	Real GDP growth				Inflation	Real GDP growth				Inflation	Real GDP growth						
	< 2%	2% to 4%	> 4%	Avg.		< 2%	2% to 4%	> 4%	Avg.		< 2%	2% to 4%	> 4%	Avg.			
	< 2%	6.9%	7.7%	5.8%		7.1%	< 2%	0.39	0.76		0.54	0.60	< 2%	1.0%	8.3%	5.5%	5.4%
	2% to 4%	2.7%	7.1%	4.5%		5.7%	2% to 4%	0.36	0.67		0.50	0.58	2% to 4%	-1.7%	6.2%	6.2%	5.2%
	> 4%	-0.9%	2.3%	1.3%		1.7%	> 4%	0.28	0.25		0.56	0.31	> 4%	4.5%	3.8%	6.2%	4.3%
	Avg.	4.3%	5.5%	4.2%		5.0%	Avg.	0.36	0.54		0.52	0.51	Avg.	0.8%	5.8%	6.0%	5.0%

Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Innovation and rising productivity growth have often resulted in prolonged periods with a favourable structural growth/inflation mix (Exhibit 18). The early 20th century was defined by the Internal Combustion Engine and Electrification, enabling the rise of mass production, urbanisation, and global logistics, resulting in the Golden 20s. The mid-century invention of the Transistor and the Microprocessor catalysed the computing revolution, automating clerical work and laying the groundwork for the Internet – leading to the Golden 50s. Since the 1990s, globally integrated trade and supply chains have become tailwinds for manufacturing in developed markets. Since the 2000s, the internet has led to the rise of platform companies, or ‘hyperscalers’, leveraging network effects to scale at near-zero marginal cost. Our economists expect AI to meaningfully boost productivity growth over the next decade through task automation in knowledge industries.

Exhibit 18: Innovation has historically often conditioned the structural growth/inflation mix
10-year rolling average of US GDP growth, CPI inflation and labour productivity growth



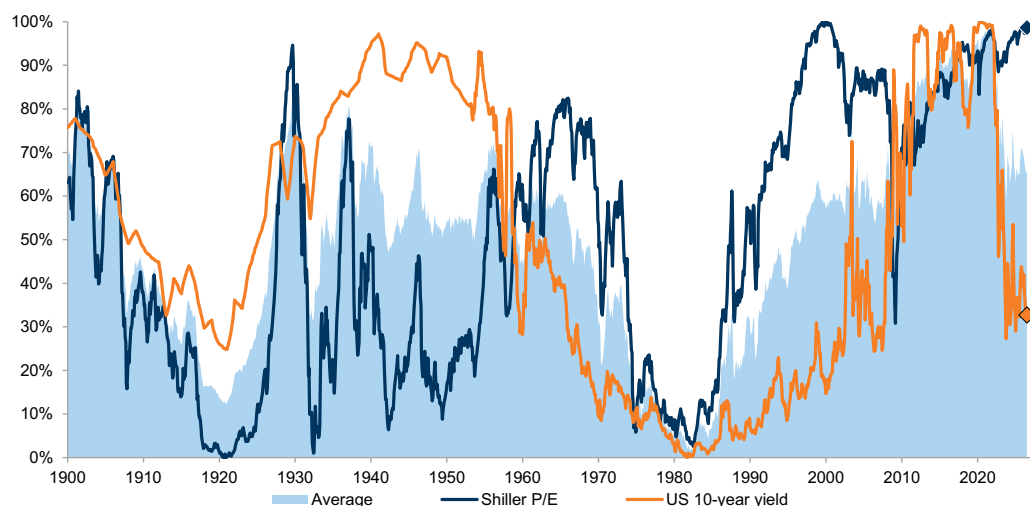
Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Innovation and valuation frustration

During such periods of innovation with a favourable growth/inflation mix, both equities and bonds tend to become more expensive (Exhibit 19). As we first highlighted in our ‘Balanced Bear’ research in 2021, this increases vulnerability to stagflationary shocks – the post COVID-19 reopening surge in inflation eventually triggered one of the largest drawdowns for 60/40 portfolios on record. Since then, cyclically-adjusted equity valuations have recovered and are back near all-time highs. However, due to continued inflation volatility, more recently linked to the Middle East war, and fiscal concerns, global bond yields have settled near their long-run averages. This points to lower prospective equity risk premia.

Exhibit 19: Equities are expensive, bond yields near long-term average

Valuation percentile (since 1871)

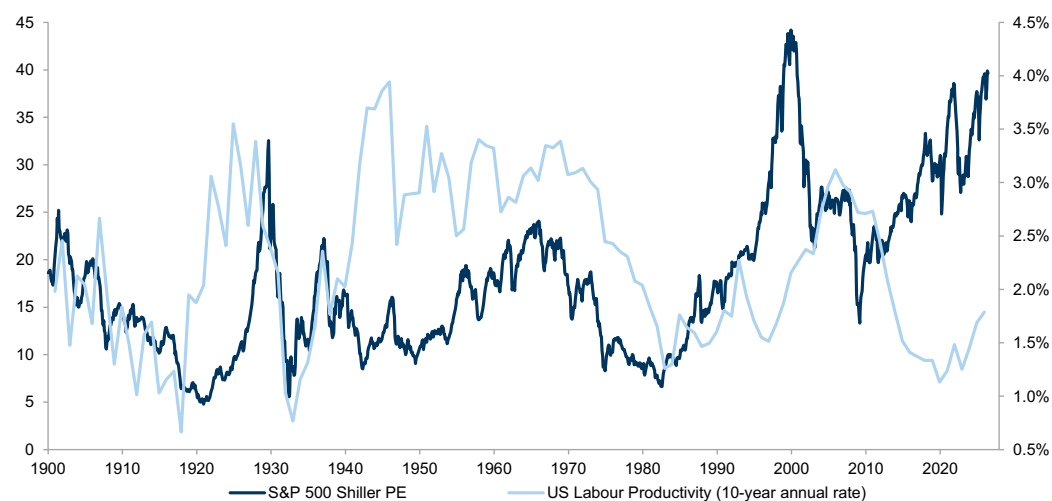


Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

Historically, S&P 500 Shiller P/E's often increased before productivity growth picked up, in part because front-loaded investments shifted earnings above their long-term trend. This can result in a boom/bust cycle in the event of concerns about overinvestment or insufficient productivity benefits from the technology. During Tech booms, there is often a pick-up in competition and new disruptive technologies, which can weigh on profitability and increase the risk of excess capacity. As a result, there is a risk of bubbles similar to those seen in the late 1990s – this has again become a key concern as Shiller P/E's have picked up materially since the GFC while productivity growth has remained subdued.

Exhibit 20: S&P 500 Shiller P/E's are nearing the Tech Bubble levels

S&P 500 Shiller P/E vs. US labour productivity growth



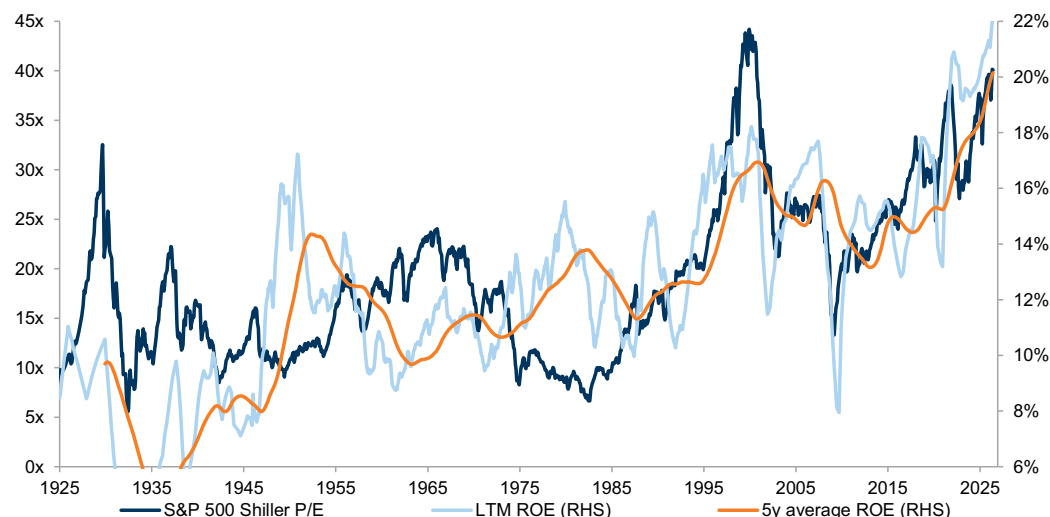
Source: Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

From spending to earning — capex vs. productivity benefits

An important difference between the current AI boom and the 2000 Tech Bubble has been high corporate profitability. Higher ROEs help explain higher S&P 500 Shiller P/E's since the 1990s – they have been boosted by rising profit margins, extending a trend that started in the early 1980s. Post GFC, US corporate profitability increased further owing to the fast-growing US Tech sector, falling interest costs, low labour cost inflation, tax cuts and share buybacks. The S&P 500 trailing 4-quarter ROE reached a record high of 22% in Q1 2026, surpassing the previous high reached in 2021.

Exhibit 21: Rising ROEs have supported US equity valuations

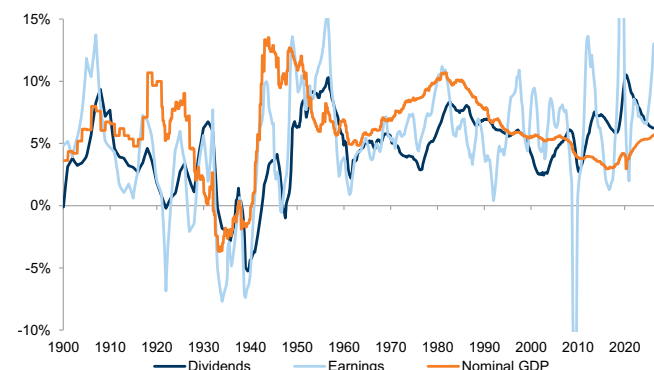
Return on equity (ROE) based on GAAP operating earnings (ex restructuring charges)



Source: Kenneth French, Robert Shiller, Haver Analytics, Goldman Sachs Global Investment Research

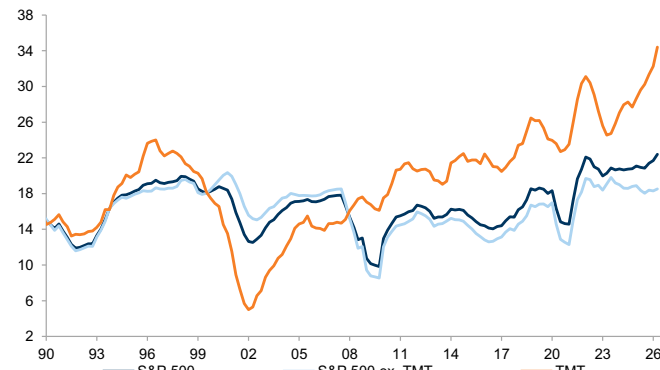
While US GDP growth was relatively low post GFC, the corporate sector materially outperformed the broader economy in the US (Exhibit 22). Similarly, the global Tech sector has been a major driver of earnings growth for global equities – in fact, since the GFC, earnings growth for the MSCI World excluding Tech has been much lower and below long-run averages. As our US strategy team has shown, the rising share of mega cap Tech stocks such as the ‘hyperscalers’, with their capital-light business models and strong revenue growth, has been a key driver of this trend (Exhibit 23).

Exhibit 22: S&P 500 earnings have accelerated in recent years and outstripped GDP growth



Source: Robert Shiller, Goldman Sachs Global Investment Research

Exhibit 23: US TMT stocks are more profitable compared to the Tech Bubble and the rest of the market



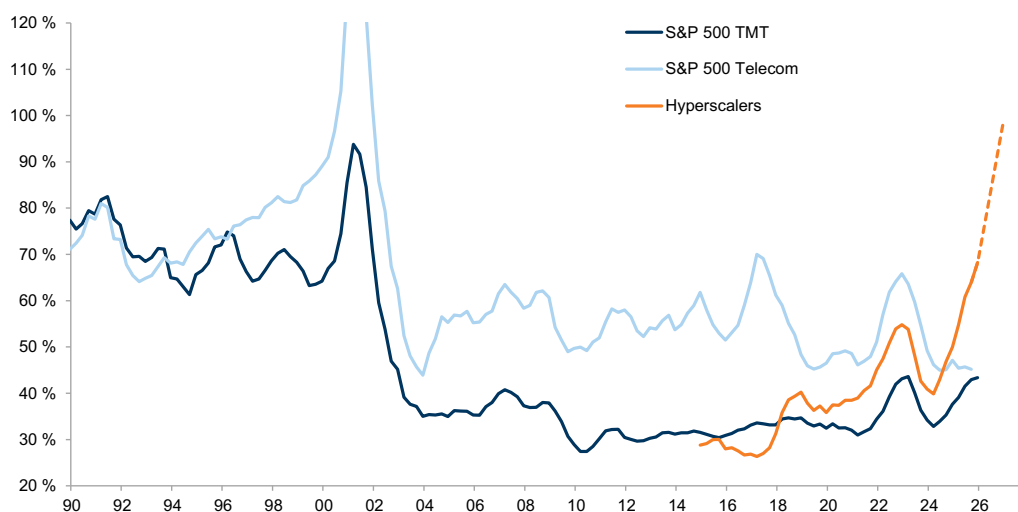
Source: Compustat, Goldman Sachs Global Investment Research

However, the ‘hyperscalers’ have now materially increased their AI capex spending.

As we wrote in Global Strategy Paper: The Post Modern Cycle – Navigating the Capex Boom, this has boosted profitability for capex beneficiaries such as semiconductor companies. The ‘hyperscalers’ are now on track to allocate all of their cash flows from operations to capex (Exhibit 24). As a result the group has sharply reduced share repurchases and begun raising additional capital via equity and debt issuance. As many of the beneficiaries, such as semiconductor companies, are highly cyclical, the risk from reductions in AI capex has increased. Our US strategy team remains optimistic about near-term momentum in AI capex. However, AI capex growth is likely to slow down eventually due to efficiency gains, competition and capital constraints.

Exhibit 24: AI capex spending from the hyperscalers has accelerated

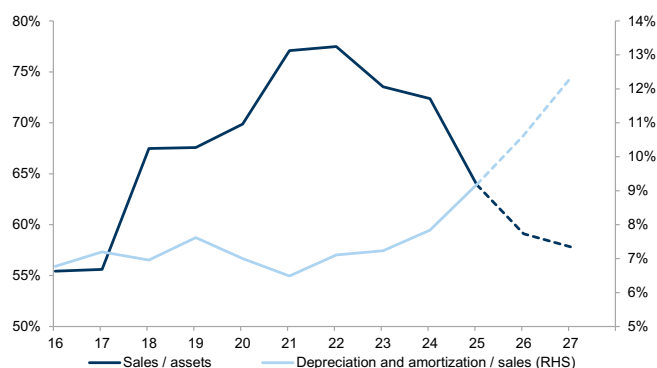
Capex as a % of cash flow from operations, dashed line represents 2026 consensus



Source: Compustat, FactSet, Goldman Sachs Global Investment Research

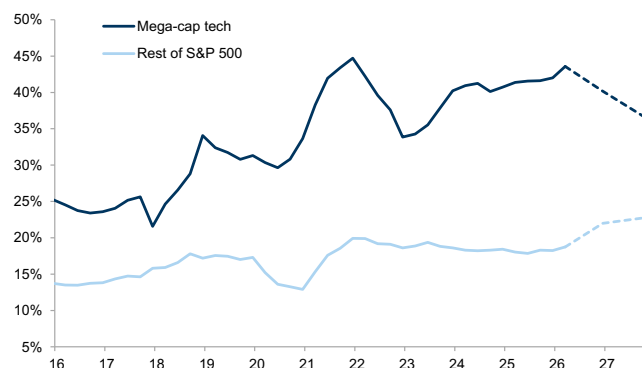
And as our US strategy team highlighted, the outlook for profitability among the hyperscalers is becoming more challenging. They are becoming increasingly asset-heavy businesses. AI capex has led to a decline in asset turnover (sales/assets), which analysts expect will continue in 2027 (Exhibit 25). Consensus estimates also suggest that each of the mega-cap Tech stocks will experience declining ROEs from next year, by an average of 7 pp (Exhibit 26). The hyperscalers and the broader AI ecosystem will have to demonstrate that they can monetise AI applications – a process that may take longer, with profitability potentially declining first.

As our economists wrote, while ICT investment began rising sharply in the early 1980s, industry-level data show that its impact on productivity followed a J-curve: a modest drag on measured productivity growth in the first four years, with clear gains emerging only after eight years and peaking in year twelve. Having said that, they expect AI to have a quicker impact on productivity because costs are falling much faster for AI models than they did for ICT equipment, and network effects are less vital to realising productivity gains. While AI adoption should support broader corporate profitability in the long run, it may also cause greater disruption to existing business models on the way, particularly in software.

Exhibit 25: AI hyperscaler capex creates headwinds to ROE

Hyperscalers = AMZN, META, MSFT, GOOGL, ORCL

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 26: ROEs for Mega-cap tech stocks are expected to decline

Mega-cap tech = AAPL, AMZN, AVGO, GOOGL, META, MSFT, NVDA

Source: Compustat, FactSet, Goldman Sachs Global Investment Research

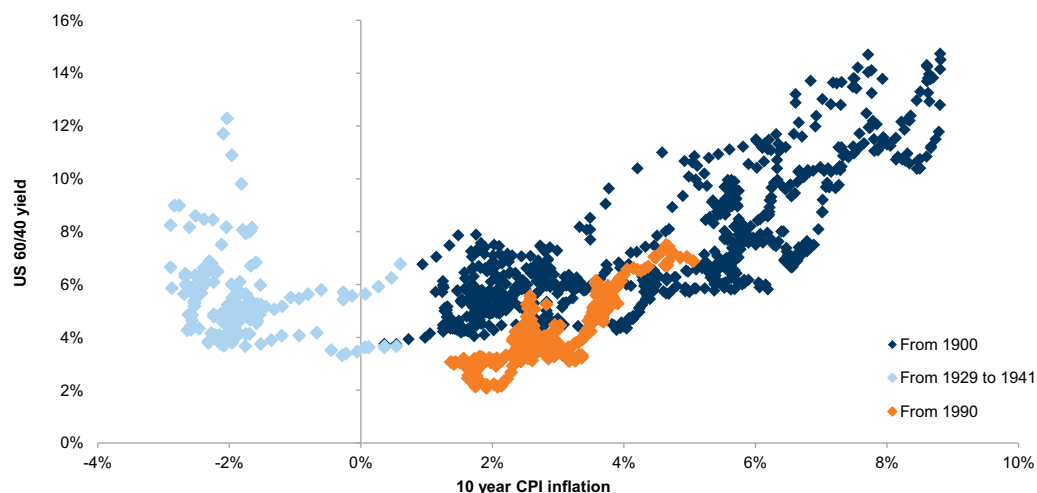
Risks from inflation for valuations

While innovation creates upside for balanced portfolios, inflation can drive

downside risks. Exhibit 27 shows that low and anchored inflation, i.e., US CPI inflation close to or slightly below 2%, has historically anchored 60/40 yields lower, while either very high or very low inflation has weighed on valuations. As we wrote before, inflation volatility and fiscal spending are likely to be higher due to a combination of adverse trends: (1) *Deglobalisation*, both economically and geopolitically, (2) *Decarbonisation*, with the risk of supply shocks and rising costs associated with climate change, and (3) *Demographics*, such as lower population growth, higher dependency ratios and income inequality. In addition, the current AI capex boom may drive further inflationary pressures – as our economists recently highlighted, the spike in memory and chip prices has boosted core PCE, albeit partly due to measurement issues.

Exhibit 27: Anchored inflation near 2% has been most supportive for valuations

US 60/40 yield = 60% S&P 500 Shiller earnings yield + 40% US 10-year yield



Source: Robert Shiller, Goldman Sachs Global Investment Research

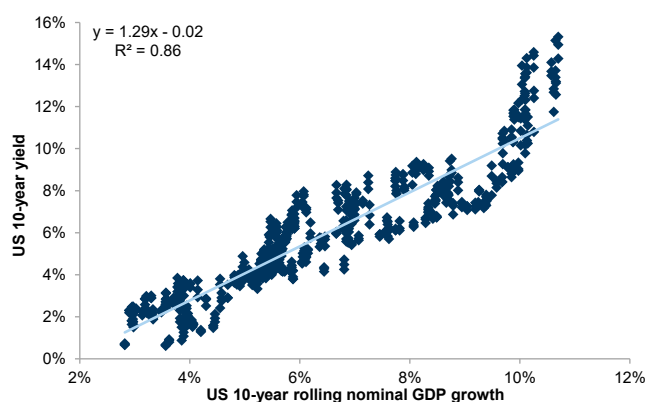
Long-dated bond yields tend to increase with rising nominal growth, even if that

growth is solely due to inflation – the 10-year US nominal GDP growth, i.e. real growth + inflation, has explained nearly 90% of the variation in US 10-year yields since WW2 ([Exhibit 28](#)). Other drivers, such as monetary and fiscal policy regimes, inflation volatility and expectations for productivity growth, can also influence long-dated bond yields. As our rates team has [highlighted](#), the AI boom could increase neutral rates by raising the expected return on capital, leading to lower savings and higher investments.

The response of equity valuations to inflation is less symmetric ([Exhibit 29](#)). A good environment for equities has been range-bound inflation between 1% and 3%, which has mostly been the case since the late 1990s. High and rising inflation (>3%), as seen during the 1970s and in 2022, weighed materially on equity valuations to concerns about profit margins and rising discount rates. However, too low inflation increases deflation fears, pushing up equity risk premia as the risk of profit stagnation and defaults increases. This was the case during the Great Depression of the 1930s, as well as in Europe and Japan during the 1990s and post-GFC periods, respectively.

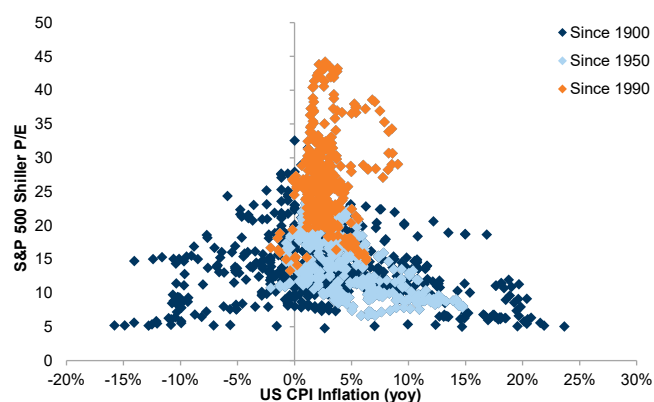
Exhibit 28: Bond yields track nominal growth

Data since 1950



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 29: Low and anchored inflation has also supported higher equity valuations



Source: Robert Shiller, Goldman Sachs Global Investment Research

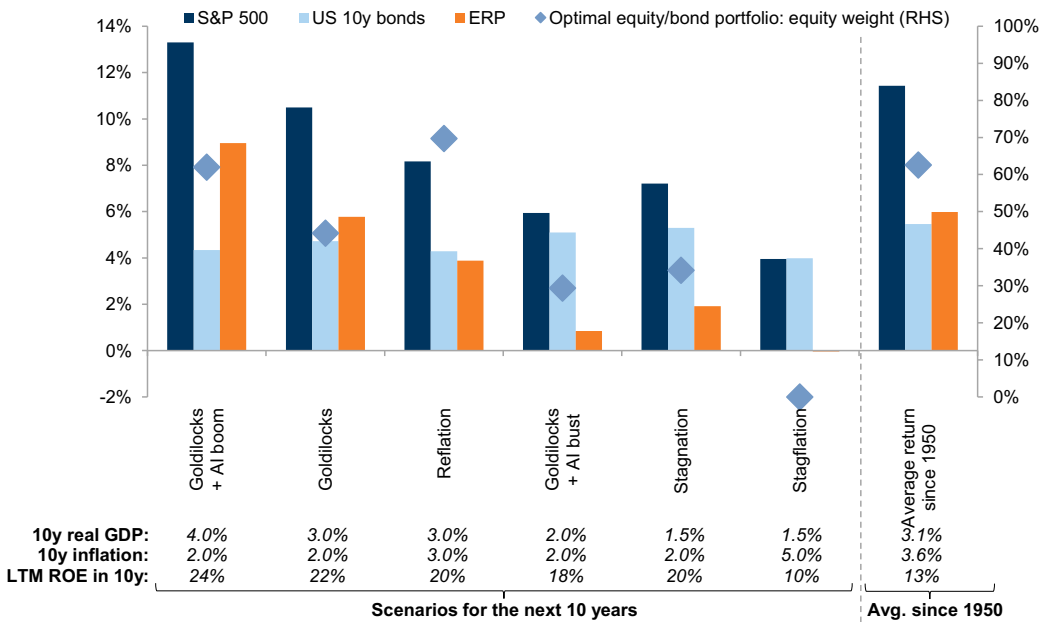
Updating our structural cycle scenarios — between AI boom and bust

The outlook for balanced portfolios in the coming years is closely linked to the AI boom and risks from inflation.

We derive new scenarios using our macro-based long-term return framework from [Global Strategy Paper: The Strategic Balanced Bear](#) ([Exhibit 30](#)): (1) Goldilocks + AI boom, with continued ROE expansion and strong GDP growth, (2) Goldilocks, but with stable ROE, (3) Reflation, with strong nominal growth but lower ROE and (4) Goldilocks + AI bust, in which ROE declines and GDP growth weakens, (5) Stagnation, with weaker growth but ROE somewhat in-line with its 5-year average and (6) stagflation, similar to the 1970s.

In all scenarios except ‘Goldilocks + AI boom’, equity returns are below the long-run average. Equity risk premia for both the ‘normal’ Goldilocks backdrop and for Reflation are near the long-run average – this still suggests equity allocations that are closer to the long-run average optimal levels of 50-60%. However, for the more negative scenarios equity returns and equity risk premia are much worse, pointing to lower equity allocations going forward. But bond returns are also somewhat below average in most scenarios and unlikely to provide much a large buffer in those adverse scenarios, especially those with inflation risk.

Exhibit 30: Structural regimes point to different long-term return forecasts
10-year return scenarios in function of different structural scenarios



Source: Haver Analytics, Robert Shiller, Goldman Sachs Global Investment Research

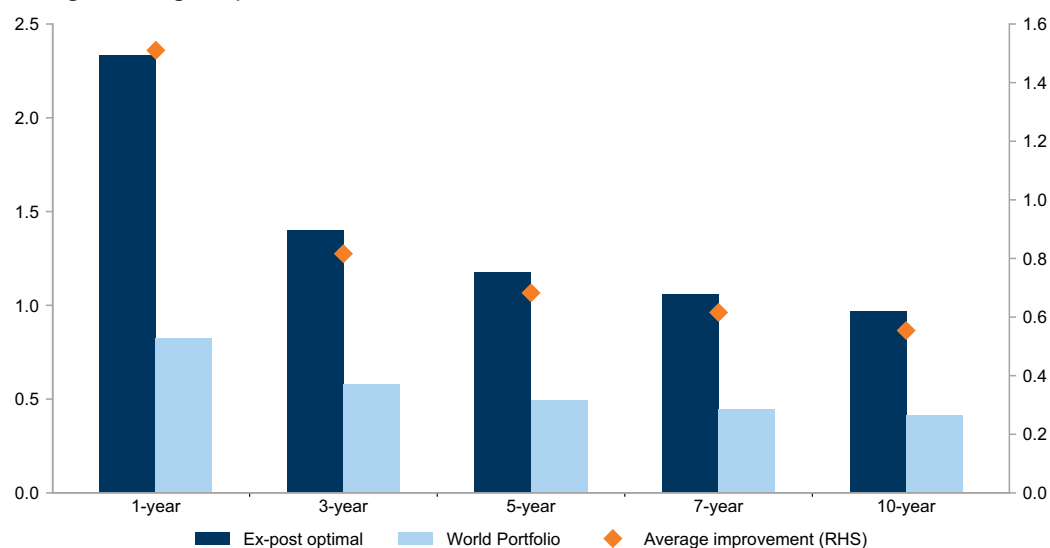
The World (Portfolio) is not enough — investigating regime-neutral portfolios

Market-value weighted benchmarks are seldom optimal

As the weights of asset classes, regions and sectors within market value-weighted benchmarks drift as a function of past performance and asset supply, they tend to be backward-looking. Optimal portfolios, in contrast, vary as a function of future macro regimes. To compare the 'World Portfolio' benchmark with what would have been mean-variance-efficient, we split it into 5 big building blocks (US and non-US equities, US and non-US bonds, and Gold) and create ex-post optimal portfolios. Unsurprisingly, over shorter horizons, for example 1-3 years, business cycle swings and bear markets drove large market timing opportunities (in hindsight) and ex-post optimal Sharpe ratios are much higher ([Exhibit 31](#)).

Exhibit 31: The World Portfolio materially lags the most efficient portfolio, even over longer horizons

Average of rolling Sharpe ratios (data since 1950)



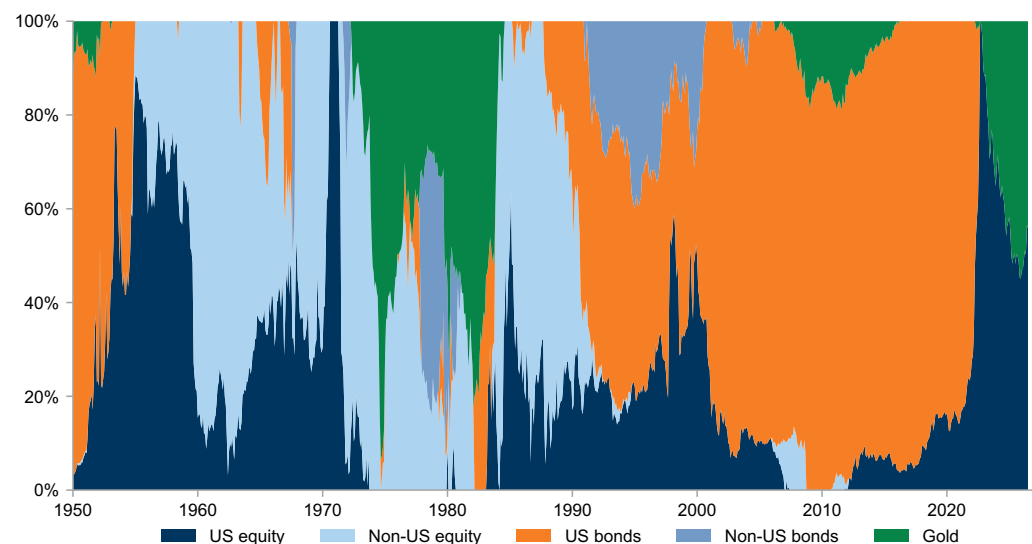
Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Even over 10-year rolling horizons, the ex-post optimal portfolio mix shifted notably over time ([Exhibit 32](#)).

As we discuss in [Global Strategy Paper: The Strategic Balanced Bear](#), structural cycles, i.e. trends in growth, inflation and policy, impact optimal multi-asset portfolios for prolonged periods of time. During the Golden 50s, the 1990s and the post-GFC period, there were larger allocations to US equities due to a boost from technological innovations. In the 1960s and 70s, the optimal portfolio had no allocations to bonds. In the 1970s stagflation Gold dominated the optimal portfolio and similarly played a more important role during the GFC and in recent years. After the GFC and before the COVID-19 crisis, a US risk parity portfolio (30-40% equities, 60-70% bonds) had the highest risk-adjusted returns. Since the COVID-19 crisis, the optimal US equity weight has increased, and more recently so has the optimal allocation to Gold. The ex-post optimal portfolio over the last decade was roughly 50% US equities and 50% Gold.

Exhibit 32: Over 10-year rolling horizons, the optimal portfolio shifted notably over time, deviating materially from the World Portfolio benchmark

10-year rolling weight of ex-post highest Sharpe ratio portfolio (data since 1950)



Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Regime-neutral portfolios tend to outperform in the medium-term

This raises the question of how to construct a portfolio that performs robustly across macro regimes. This is important to assess and manage drifts in market value-weighted benchmark weights and realistically outperform the 'World Portfolio'. We construct 4 simple, regularly rebalanced portfolios that aim to be 'regime-neutral'. We define a 'regime-neutral' portfolio as one whose weights do not rely on any single macro regime so that it performs well through cycle and is identifiable ex ante. In practice, we approximate this concept using different approaches:

- 1. The average of the 10-year rolling ex-post optimal portfolio weights since 1950** (from [Exhibit 32](#)). This approach aims to average out variation in optimal weights due to different macro regimes to create a 'neutral' starting point.² It incorporates historically changing risk/rewards and correlations across assets.
- 2. The average of the historical weights of the 'World Portfolio' benchmark since 1950**, assuming they represent the long-run equilibrium. Average asset weights since 1950 have been 25% US equity, 24% non-US equity, 30% US bonds, 18% non-US bonds and 4% Gold.
- 3. A risk-parity strategy**, which allocates to assets based on their risk to reduce risk concentration towards equities that is common in balanced portfolios. There is no need for return or correlation assumptions. We use a simple approach that weights assets based on the inverse of their volatility of monthly returns since 1950.
- 4. Naïve 1/N weights, i.e. equal weights**, as a robustness check to address the concerns about out-of-sample performance of optimal portfolios raised by DeMiguel et.al. (2009).

² This is similar to the idea of constructing a resampled efficient frontier from Michaud (1998), who uses Monte Carlo resamples of the inputs to reduce estimation error rather than averaging optimal portfolio weights across historical windows.

All regularly rebalanced ‘regime-neutral’ approaches outperformed the ‘World Portfolio’ on a risk-adjusted basis since 1950 (Exhibit 33). They mostly had better hit ratios of generating attractive real returns over the long run while reducing the risk of negative real returns. Compared to the current ‘World Portfolio’ weights, the ‘regime-neutral’ portfolios generally point to somewhat smaller weights in equities (and US equities in particular), larger weights to Gold and smaller weights to non-US bonds.

The risk-parity strategy had the lowest Sharpe ratio, while the one based on 10-year rolling average optimal weights offered the highest Sharpe ratios. While the ‘regime-neutral’ portfolios are estimated ex post, the persistent improvement observed over long horizons suggests that regime diversification, rather than timing, is the key driver. Even the naïve 1/N strategy outperformed the ‘World Portfolio’ benchmark, indicating that the results are not solely dependent on optimisation.

Exhibit 33: A ‘neutral’ global multi-asset benchmark would be very different to the current World Portfolio (for a US\$ investor)

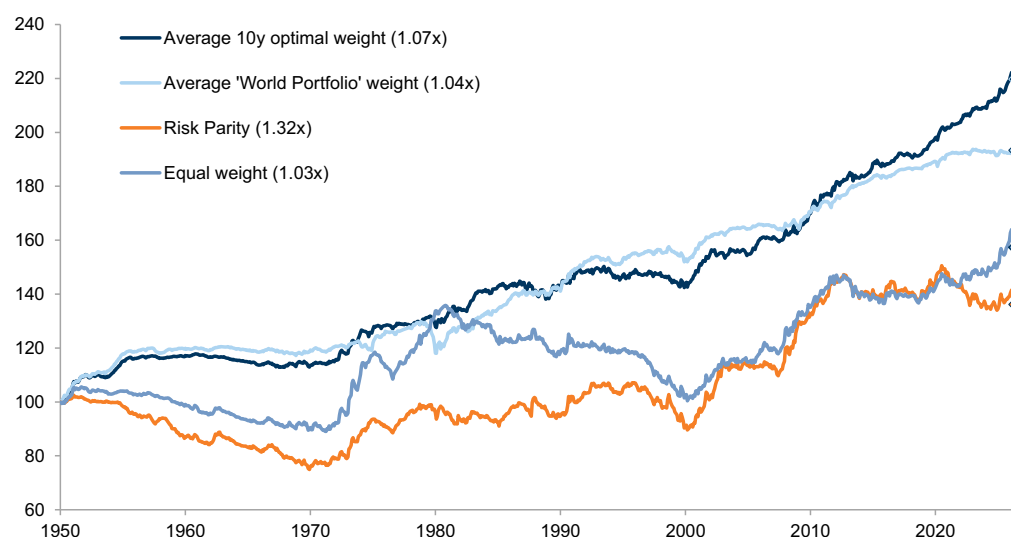
	World Portfolio	Average 10y optimal weight	Average ‘World Portfolio’ weight	Risk Parity	Equal weight
Asset weights					
	Current				
US equity	35%	24%	25%	13%	20%
Non-US equity	20%	23%	24%	13%	20%
US bonds	19%	38%	30%	40%	20%
Non-US bonds	19%	5%	18%	23%	20%
Gold	6%	11%	4%	11%	20%
Performance metrics since 1950					
Annualised return	7.9%	8.7%	8.7%	7.4%	8.4%
Annualised volatility	8.2%	7.6%	7.8%	6.2%	7.9%
Tracking error		1.7%	1.4%	3.3%	2.8%
Sharpe ratio	0.45	0.58	0.56	0.51	0.53
Information ratio		0.44	0.49	-0.16	0.17
5% CVaR of 1m returns	-5.0%	-4.5%	-4.7%	-3.4%	-4.5%
1y relative return	5th perc.	-1.9%	-1.4%	-6.2%	-4.2%
	95th perc.	4.1%	4.0%	7.4%	6.6%
10y relative return	5th perc.	-0.4%	-0.1%	-2.6%	-1.5%
	95th perc.	1.6%	1.5%	1.8%	3.2%
Hit ratio for real returns since 1950					
1y rolling real return	above 5%	50%	54%	42%	48%
	below 0%	24%	22%	23%	25%
10y rolling real return	above 5%	35%	42%	26%	37%
	below 0%	5%	1%	3%	0%

Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Leaning against momentum can be costly in the short-term
While the benefits of regime diversification accrue over the long-run, benchmark drift can dominate relative performance in the short-term. Comparing relative performance on a risk-neutral basis shows prolonged periods (usually during strong equity bull markets) when the different ‘regime-neutral’ portfolios underperformed (Exhibit 34). In those periods benchmark drifts can exacerbate positive momentum. For example, during the Tech Bubble, all ‘regime-neutral’ portfolios struggled to keep up with the ‘World Portfolio’, which had a growing equity weight. This was particularly true for the risk parity-based and equal-weight portfolios due to their lower equity weight. The same was true during the Golden 1950s and 1960s. This illustrates the difficulty of leaning against benchmark drifts in equities, especially during strong bull markets. Approaches to better time when to lean against large benchmark drifts could be based on ‘regime-based’ asset allocation signals (see [Global Strategy Paper: The Dynamic Balanced Bear](#)) or momentum-based signals and volatility targeting overlays (see [Global Strategy Paper: Balanced Bear Despair – Part 4](#)).

Exhibit 34: Regime-neutral portfolios outperformed in the medium-term

Relative return vs World Portfolio (risk-neutral scaling to match the World Portfolio vol, in brackets)

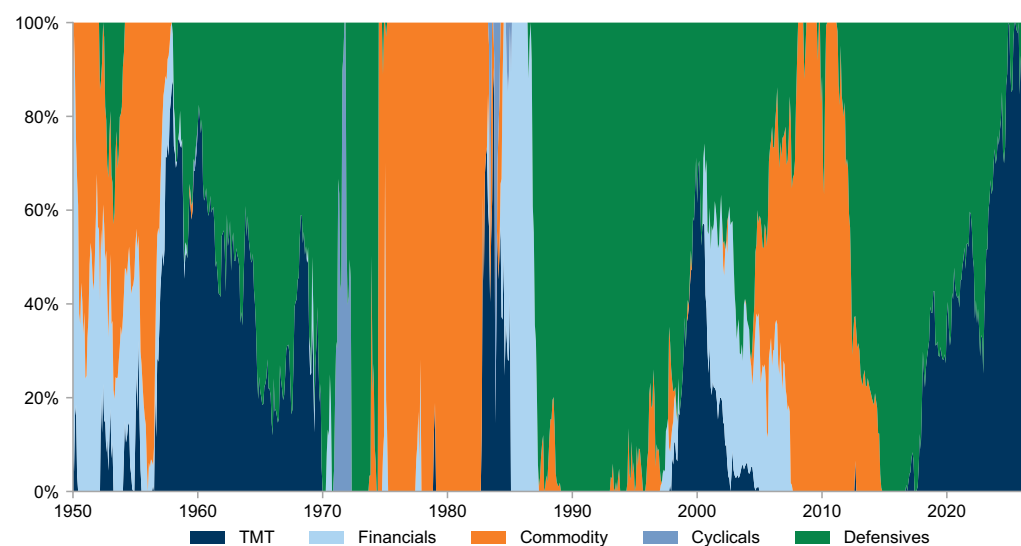


Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Leaning against sector drifts within equity benchmarks is especially difficult. As shown in [Exhibit 35](#), over 10-year rolling horizons the ex-post optimal sector mix within a US equity benchmark shifted notably over time. For example during Technology booms like the 1950s, late 1990s and more recently, there have been material tilts towards the Technology sector. The 'regime-neutral' Technology sector weight (c.18%) is well below current levels within US equities (c.47%). During periods of strong momentum in the Technology sector, it can be costly to lean against benchmark drifts and rebalance back to 'regime-neutral' weights. With historical high volatility of the Technology sector, especially during booms, investors need to balance near-term tracking error risk from reducing the Technology sector weight with potential long-term improvements in the risk/reward.

Exhibit 35: Optimal sector weights varied materially over time

10-year rolling optimal sector weights for US equities



Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

Benefits from broadening real asset allocations

The vulnerability of balanced equity/bond portfolios to higher inflation, especially stagflation regimes, points to benefits from larger allocations to real assets. ‘Real assets’ also provided diversification in Tech-led equity drawdowns. Due to size and liquidity constraints we included only Gold in our ‘World Portfolio’ benchmark. Over the long run, Gold had a low Sharpe ratio but occasionally, like in recent years or the 1970s stagflation, it delivered very strong returns and diversification for balanced equity/bond portfolios. But Gold is an incomplete inflation hedge; while it protected purchasing power over the long run, it had a mixed relationship with inflation over shorter horizons – in recent years it has performed well mainly due to concerns about currency debasement and institutional credibility.

Adding a broader set of real assets can better diversify different sources of inflation. If inflation is demand-driven, companies with pricing power and cyclical commodities benefit, while cost-push inflation due to energy supply shocks favours related commodities. While real asset stocks (infrastructure/real estate/energy) cannot provide an imminent inflation hedge they protect from inflation in the medium-term and have higher Sharpe ratios. We include the equal-weight index of Gold, S&P GSCI and different real asset stocks – as [Exhibit 36](#) shows, compared to Gold alone this ‘Real asset index’ received a larger weight in a ‘regime-neutral’ portfolio since 1950. While the World Portfolio is currently very exposed to innovation, it might not have enough protection from inflation, pointing to higher allocations to real assets.

Exhibit 36: Benefits from broader diversification beyond market value weights

		Average 10-year optimal weight adding:				
		World Portfolio	Gold	S&P GSCI	Real Asset Stocks	Real Asset Index
Asset weights		<i>Current</i>				
US equity		36%	24%	24%	18%	21%
Non-US equity		20%	23%	24%	23%	18%
US bonds		19%	38%	37%	37%	36%
Non-US bonds		19%	5%	4%	6%	5%
Gold		6%	10%	-	-	-
S&P GSCI		-	-	10%	-	-
Real Asset Stocks		-	-	-	16%	-
Real Asset Index		-	-	-	-	20%
Performance metrics since 1950						
Annualised return		7.9%	8.7%	8.7%	9.2%	9.0%
Annualised volatility		8.2%	7.6%	7.8%	8.3%	7.8%
Tracking error			1.6%	2.7%	2.6%	2.4%
Sharpe ratio		0.45	0.58	0.57	0.59	0.61
Information ratio			0.44	0.27	0.47	0.41
5% CVaR of 1m returns		-5.0%	-4.5%	-4.8%	-5.1%	-4.7%
1y relative return	5th perc.		-1.9%	-4.1%	-3.5%	-3.5%
	95th perc.		4.1%	6.1%	6.8%	5.9%
10y relative return	5th perc.		-0.4%	-0.4%	0.0%	0.2%
	95th perc.		1.5%	2.1%	2.4%	1.8%
Hit ratio for real returns since 1950						
1y rolling real return	above 5%	50%	54%	55%	58%	56%
	below 0%	25%	22%	24%	22%	20%
10y rolling real return	above 5%	35%	42%	40%	55%	48%
	below 0%	5%	1%	2%	3%	1%

Source: Bloomberg, Datastream, Haver Analytics, World Gold Council, Goldman Sachs Global Investment Research

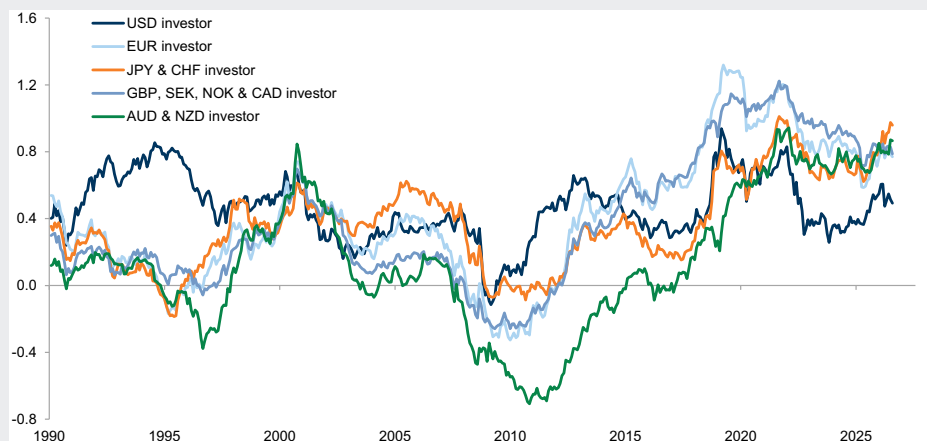
The impact of FX and the investor currency

As US weights in global equity and bond benchmarks have risen, the Dollar has become a more important driver of risk-adjusted returns for non-US investors ([Exhibit 37](#)). The impact of FX risk in portfolios depends on foreign asset exposure, home-currency volatility relative to foreign-currency exposure, and correlations

with equities. The Dollar was a drag for non-US investors in the early 1990s and into the GFC, but has generally boosted Sharpe ratios since then, supported by its uptrend and negative correlation with risky assets. This was especially true for investors with cyclical currencies such as the Australian Dollar, Swedish Krona and Canadian Dollar, while investors with safe-haven currencies such as the Swiss Franc and Japanese Yen generally saw lower Sharpe ratios up until the COVID-19 crisis.

Exhibit 37: Sharpe ratios also depend on the investor FX

Average of 10-year rolling Sharpe ratios of the World Portfolio

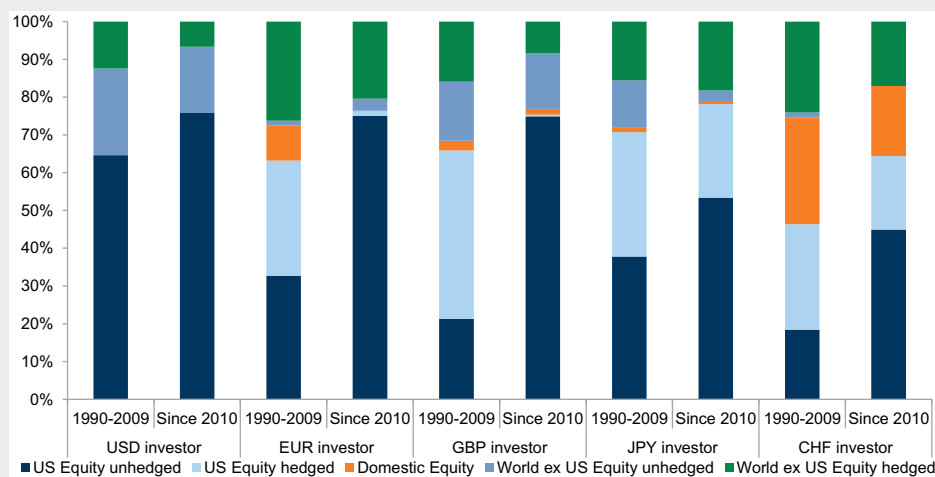


Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Both optimal regional equity allocations and FX hedge ratios vary by investor currency. We split global equities into five buckets: domestic equities, US equities hedged to domestic FX, US equities unhedged, World ex US equities unhedged and World ex US equities hedged. We then look at average 10-year rolling optimal weights ([Exhibit 38](#)): Since 2010, large unhedged US equity weights benefited most investors, though less so before the Dollar strengthened. Hedge ratios and allocations also differed by currency: Swiss investors, for example, generally required higher FX hedge ratios and larger domestic equity allocations because of the Swiss Franc's strength and safe-haven characteristics. This highlights both the benefits and challenges of incorporating FX in long-horizon portfolio construction.

Exhibit 38: FX risk can still impact optimal regional allocations

Average weights for optimal regional equity allocations: 1990–2009 versus 2010 to present



Source: Bloomberg, Datastream, Haver Analytics, Goldman Sachs Global Investment Research

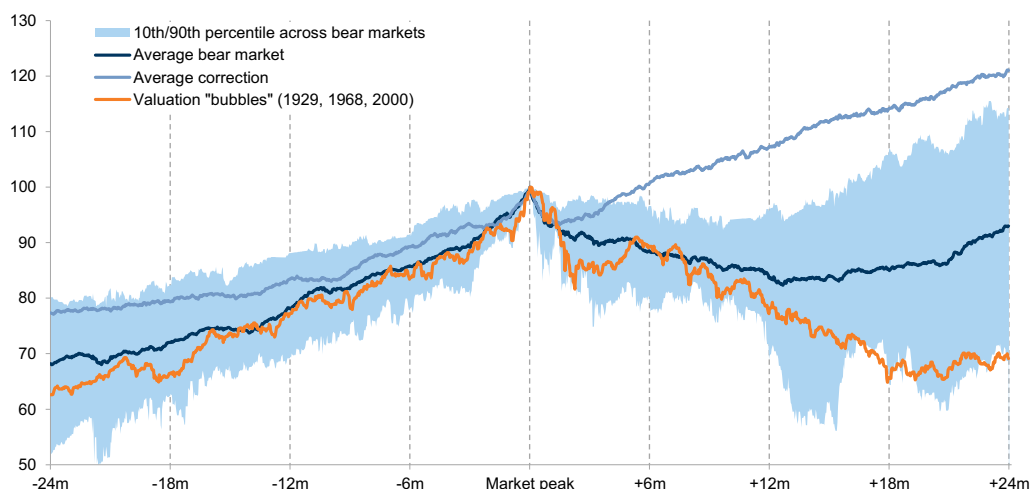
Finding the right balance between Innovation and Inflation

Late cycle investing — time in the market vs. timing the market

Even though there is a case for leaning against the drift to equities and Technology stocks in the medium-term, being underinvested can be costly in the short-term, with equities usually performing well up until the bull market peak. Avoiding large S&P 500 drawdowns would unsurprisingly boost performance but missing positive returns would have been very costly. Thus, timing the market always competes with time in the market. Corrections tend to be short-lived and thus difficult to time. And while bear markets are deeper and longer, the average profile since 1900 shows that investors that divested 6 months before the peak roughly lost as much as those that were 6 months late ([Exhibit 39](#)). It is more important to avoid large losses after the start of a bear market.

Exhibit 39: Equities tend to perform well into bull market peaks

Average S&P 500 total return performance around bear markets and corrections (data since 1900, 21 bear markets, 30 corrections included)

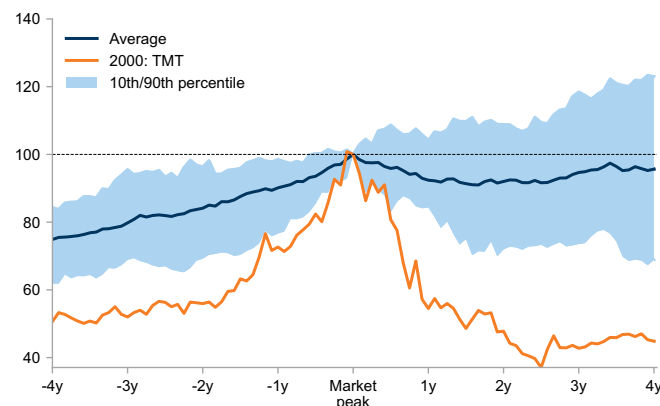


Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Leading sectors also tend to continue outperforming into the bull market peak

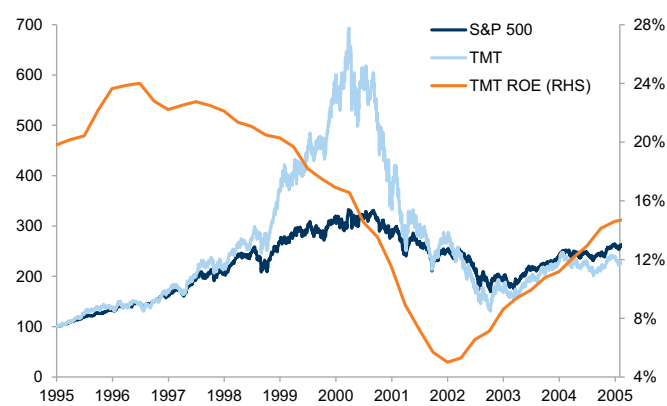
([Exhibit 40](#)). As mentioned earlier, this was particularly pronounced during Tech booms like in the 1950s and 60s as well as the late 1990s – this makes leaning against sector momentum particularly costly. This is particularly true when leading sectors are being driven by earnings growth, as is the case today or was the case for Financials leading into the GFC. However, during Tech booms, price momentum can continue without fundamentals – during the Tech Bubble, Tech stocks continued to perform until 2000 although profitability for the TMT sector peaked around 1996 ([Exhibit 41](#)).

Exhibit 40: Best performing sectors do well up until peak
Average perf. leading sector vs. the market (data since 1950)



Source: Datastream, Goldman Sachs Global Investment Research

Exhibit 41: In the Tech Bubble the ROE peaked before the market



Source: Compustat, Goldman Sachs Global Investment Research

Five investment ideas to balance innovation and inflation in portfolios

Investors need to keep exposure to innovation, protect against inflation and improve risk mitigation within portfolios.

While a growing Tech sector in benchmarks and portfolios makes portfolios imbalanced and more risky, leaning against a Tech boom is difficult. In most scenarios, long-term equity returns might be below long-run average but until momentum for the Tech sector more clearly moderates it is difficult to lean against it. We investigate five different risk mitigation strategies to help investors stay invested and create better balance in multi-asset portfolios.

(1) Keeping it real — selectively increasing real asset exposure

Allocations to real assets help create more balance in multi-asset portfolios in case of inflation risks, and they improve the potential for harvesting real returns.

Historically, during some of the biggest Tech sell-offs, real assets also proved defensive – on average our equal-weight ‘Real Asset Index’ delivered positive returns during TMT sector drawdowns, particularly during the Tech Bubble burst ([Exhibit 42](#)). ‘Real Asset stocks’, i.e. an equal-weight index of infrastructure, real estate and energy stocks, generally outperformed during those Tech drawdowns and helped diversify equities.

Exhibit 42: Real assets have done well during Tech-led equity drawdowns

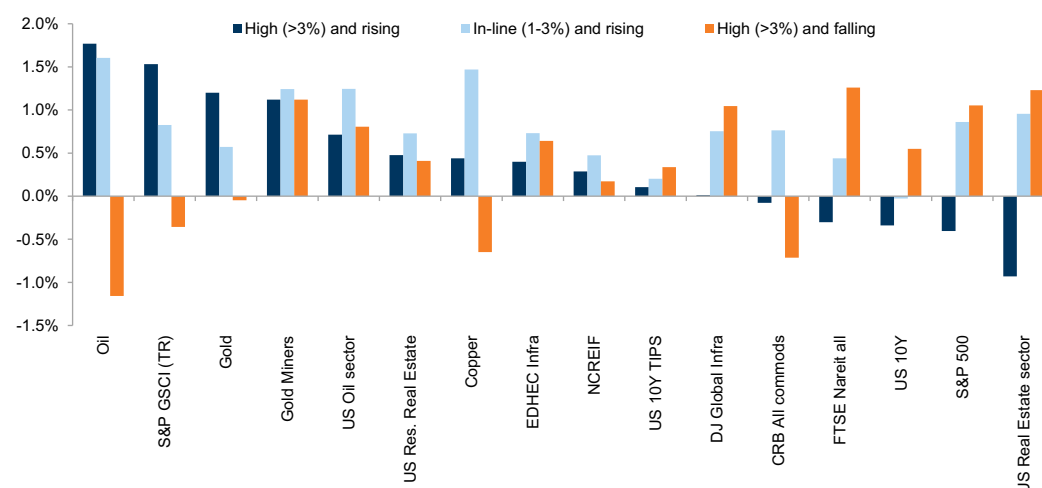
Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	Gold	S&P GSCI	Real Asset Stocks	Real Asset Index
May-50	Jul-50	2	-13%	-5%	0%		1%	-4%	
Aug-60	Oct-60	2	-10%	-6%	0%	3%	1%	-4%	-2%
Nov-61	Jun-62	7	-38%	-23%	2%	0%	-2%	-21%	-13%
Dec-67	Feb-68	2	-13%	-7%	2%	1%	3%	-4%	-2%
May-68	Jun-70	25	-39%	-26%	0%	-17%	18%	-18%	-10%
Jan-73	Sep-74	20	-52%	-45%	2%	129%	132%	-43%	5%
Aug-78	Oct-78	2	-14%	-10%	-2%	16%	7%	-8%	-1%
Jan-80	Mar-80	2	-16%	-11%	-7%	-26%	-15%	-11%	-15%
Jun-83	Jul-84	13	-22%	-10%	2%	-19%	5%	-4%	-5%
May-86	Sep-86	4	-12%	-6%	7%	24%	11%	4%	9%
Aug-87	Aug-90	36	-37%	-2%	31%	-15%	114%	13%	24%
Mar-00	Sep-02	30	-79%	-46%	34%	16%	16%	1%	8%
Dec-21	Sep-22	9	-33%	-25%	-17%	-8%	22%	-6%	-1%
Dec-24	Mar-25	3	-11%	-5%	4%	19%	5%	7%	9%
Oct-25	Mar-26	5	-11%	-5%	0%	16%	39%	18%	22%
Average		10.8	-26.7%	-15.5%	3.9%	9.8%	23.7%	-5.4%	2.0%
Median		5.0	-15.8%	-9.8%	1.8%	2.1%	6.7%	-4.4%	-0.8%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

With the energy shock from the Middle East war reversing and a sharp decline in oil prices, the benefits from different real assets are changing. During supply shocks linked to energy, direct commodity exposure, i.e. GSCI and oil exposures, as well as short-dated breakeven inflation are the main hedges, especially when central banks are fighting inflation. If central banks are not fighting inflation and there are concerns about institutional credibility and currency debasement, Gold can usually provide diversification. As our commodities team recently reiterated, there is still a case for diversification into commodities – commodities that are more demand-driven such as those linked to decarbonisation, EVs and renewable energy, as well as AI infrastructure might take over from energy. While commodities fare less well when inflation is normalising, real estate and infrastructure tend to perform best and would get further support if central banks are shifting less hawkish.

Exhibit 43: Commodities have tended to outperform in periods of high and rising inflation but with falling inflation real estate and infrastructure stocks have tended to perform better
Average monthly, real return (data since 1950, inflation based on US CPI)



Note: The FTSE NAREIT All index tracks all US REITs since 1972; before that, we use the Kenneth French real estate sector. We construct the Residential Real Estate index using Case/Shiller HPI from 1987 (FHFA before) and an estimate of rental yield before transaction cost (Zillow since 2016, macrohistory database before 2016). The NCREIF index (NPI) is a quarterly, unleveraged total return for private commercial real estate properties held for investment purposes only. The Dow Jones (DJ) Brookfield Infrastructure index tracks listed companies since 2012 - before we aggregate sectors from Worldscope (1973-2012) and Kenneth French (1926-73). The EDHEC Infrastructure 300 index replicates broad market exposure to unlisted infrastructure investments gross of fees or other costs - the index is calculated monthly as total return index.

Source: Datastream, Haver Analytics, Kenneth French, Goldman Sachs Global Investment Research

(2) Don't lose Tech momentum, diversify it through style

Diversifying across styles within equities can help manage risks around Tech stock momentum. During previous TMT sector drawdowns, we found that low volatility, high dividend yield stocks outperformed and there were large rotations from growth to value (Exhibit 44). An extreme example of the diversification benefits was the Tech Bubble burst when low volatility and high dividend yield stocks were **up** with S&P 500 down 46% and the TMT sector **down** 79%. This is in part due to them benefitting from lower bond yields but even during the 2022 drawdown without large decline in bond yields they managed to outperform materially. While cyclicals did not underperform much during TMT drawdowns, there was some benefit from defensive exposures.

The ratio of momentum vs. low vol stocks nearing Tech Bubble peaks best illustrates

the risk of a reversal and need for style diversification (Exhibit 45). The recent leg in the momentum rally has been led by global Tech hardware stocks, especially semiconductors, leading the market higher. As a result, index concentration has increased in both the S&P 500 and MSCI Asia ex Japan. This points to potential for more dispersion and increasing benefits of diversification within the AI ecosystem. In addition with concerns about AI and disruption of capital-light business models there is potential for more rotation towards more capital-heavy sectors, which also fits with allocations to ‘real assets’ – our European equity strategy team has dubbed this the HALO effect: Heavy Assets, Low Obsolescence, and we also have similar baskets in the US, Europe, Asia, EM and Japan (Exhibit 46).

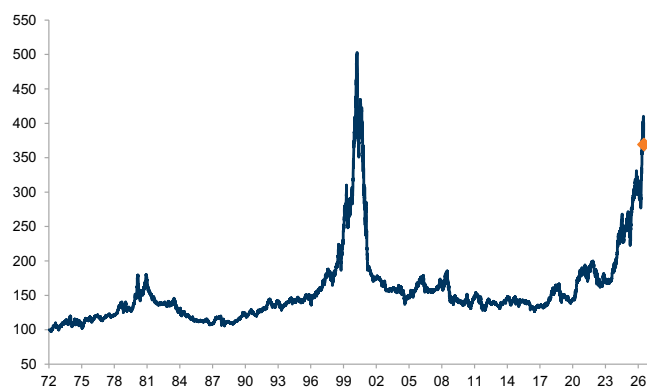
Exhibit 44: Low vol, high dividend yield, value stocks outperform in TMT drawdowns
Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	S&P low vol	High DY	Value vs Growth	Cycl vs Defs
May-50	Jul-50	2	-13%	-5%	0%		-4%	16%	5%
Aug-60	Oct-60	2	-10%	-6%	0%		-5%	2%	-3%
Nov-61	Jun-62	7	-38%	-23%	2%		-9%	12%	-4%
Dec-67	Feb-68	2	-13%	-7%	2%		-1%	6%	-3%
May-68	Jun-70	25	-39%	-26%	0%		-12%	7%	-4%
Jan-73	Sep-74	20	-52%	-45%	2%		-31%	43%	10%
Aug-78	Oct-78	2	-14%	-10%	-2%		-6%	0%	0%
Jan-80	Mar-80	2	-16%	-11%	-7%	-10%	-8%	-1%	-3%
Jun-83	Jul-84	13	-22%	-10%	2%	0%	6%	25%	-9%
May-86	Sep-86	4	-12%	-6%	7%	-1%	5%	14%	-2%
Aug-87	Aug-90	36	-37%	-2%	31%	1%	15%	-2%	-23%
Mar-00	Sep-02	30	-79%	-46%	34%	16%	30%	79%	-19%
Dec-21	Sep-22	9	-33%	-25%	-17%	-16%	-4%	22%	-5%
Dec-24	Mar-25	3	-11%	-5%	4%	7%	7%	16%	-5%
Oct-25	Mar-26	5	-11%	-5%	0%	4%	17%	17%	5%
Average		10.8	-26.7%	-15.5%	3.9%	0.1%	-0.1%	17.0%	-4.0%
Median		5.0	-15.8%	-9.8%	1.8%	0.8%	-4.1%	14.1%	-3.2%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 45: The ratio of S&P 500 momentum vs. low volatility stocks is nearing Tech Bubble peaks

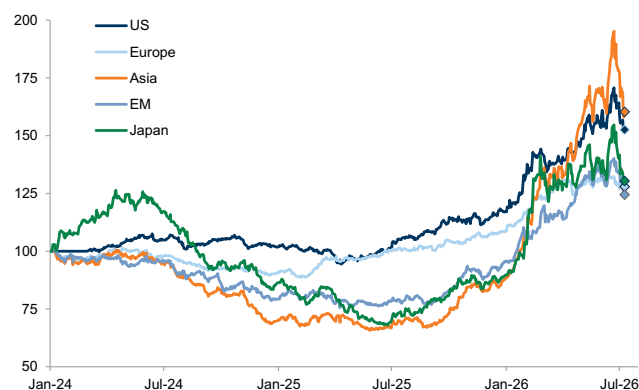
Relative performance S&P 500 momentum vs. low volatility stocks



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 46: Capital intensity has been a new driver of material dispersion within equities

Relative performance of high vs. low capital intensity baskets



Source: Bloomberg, Goldman Sachs Global Investment Research

(3) Don't get lost in translation - managing US asset dominance

We expect continued benefits from regional diversification and managing Dollar risk. Since last year, there has been less clear US asset outperformance (see also Global Strategy Paper: Magnificent Diversification). Since the start of 2025, the S&P 500 has lagged major global equity indices, in both local currency and US Dollar terms. Historically, during Tech-led equity sell-offs, US equities have on average underperformed other markets because they are more closely linked to innovation. FX

performance has been more mixed, but during the Tech Bubble burst the Dollar weakened, especially in 2002. The Swiss Franc has proven particularly defensive, although recent SNB interventions have changed that 'risk-off' characteristic. As our FX team recently highlighted, the Dollar and Tech sector performance have become more closely linked. A US equity market sell-off with clearer RoW outperformance would likely be a catalyst for Dollar depreciation.

Exhibit 47: Non-US equities tend to outperform during Tech-led equity sell-offs but FX performance is mixed

Performance during TMT drawdowns of more than 10% and 5% underperformance vs. the S&P 500

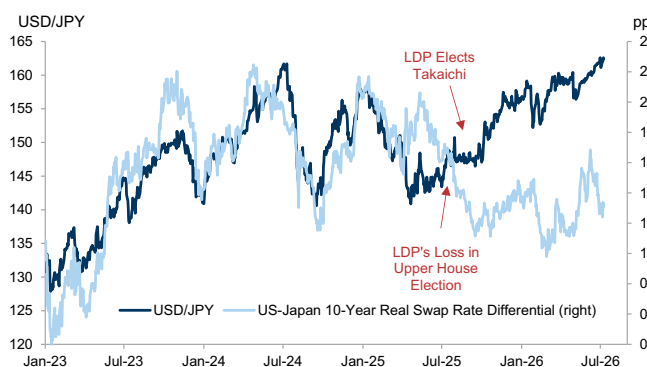
Peak	Trough	Length (months)	TMT drawdown	S&P 500	US 10y	CDAX	TOPIX	MSCI EM	CHF	EUR	JPY
May-50	Jul-50	2	-13%	-5%	0%	2%	4%	-2%			
Aug-60	Oct-60	2	-10%	-6%	0%	-6%	-	2%			
Nov-61	Jun-62	7	-38%	-23%	2%	-25%	6%	-2%			
Dec-67	Feb-68	2	-13%	-7%	2%	6%	4%	9%			
May-68	Jun-70	25	-39%	-26%	0%	-9%	42%	18%	0%	2%	0%
Jan-73	Sep-74	20	-52%	-45%	2%	-30%	-30%	-16%	22%	6%	2%
Aug-78	Oct-78	2	-14%	-10%	-2%	0%	3%	0%	9%	5%	6%
Jan-80	Mar-80	2	-16%	-11%	-7%	-5%	-3%	1%	-12%	-9%	-4%
Jun-83	Jul-84	13	-22%	-10%	2%	2%	16%	5%	-15%	-13%	-2%
May-86	Sep-86	4	-12%	-6%	7%	1%	17%	-5%	17%	10%	13%
Aug-87	Aug-90	36	-37%	-2%	31%	7%	-8%	57%	15%	15%	-1%
Mar-00	Sep-02	30	-79%	-46%	34%	-64%	-46%	-47%	13%	3%	-16%
Dec-21	Sep-22	9	-33%	-25%	-17%	-30%	-8%	-29%	-7%	-14%	-20%
Dec-24	Mar-25	3	-11%	-5%	4%	11%	-5%	2%	2%	4%	5%
Oct-25	Mar-26	5	-11%	-5%	0%	-4%	5%	0%	0%	0%	-3%
Average		10.8	-26.7%	-15.5%	3.9%	-9.6%	-0.2%	-0.4%	4.0%	0.8%	-1.9%
Median		5.0	-15.8%	-9.8%	1.8%	-4.0%	3.4%	-0.3%	2.4%	3.2%	-0.9%

Source: Datastream, Haver Analytics, Goldman Sachs Global Investment Research

In the event of a US recession, JGBs and EM bonds might provide more diversification within global multi-asset portfolios.

The Yen recently traded at its weakest level versus the US Dollar in 40 years, while JGB yields are nearing multi-year highs. As our FX team recently showed, the USD/JPY has recently decoupled from real rate differentials due to rising JGB term premia (Exhibit 48). However, the Yen often outperforms when markets are pricing in a US recession, with Fed cuts and both equity and real rates lower. While near-term our markets team expects the weaker Yen/ higher JGB yield trend to extend, eventually it might present an attractive hedge for a Tech-led US equity drawdown. Similarly, EM assets, which are negatively correlated with the Dollar, can diversify US asset-heavy portfolios (Exhibit 49) - our EM team has also found that EM assets have become more resilient in 'risk off' episodes.

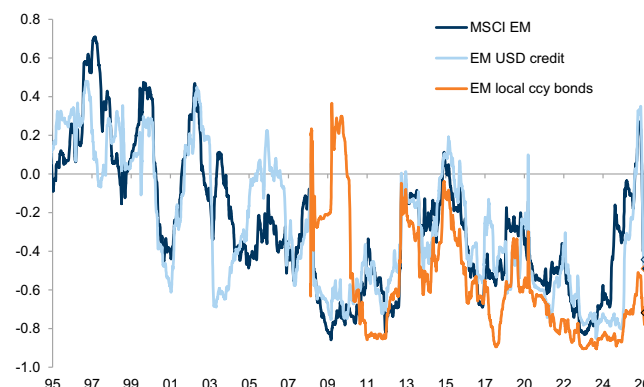
Exhibit 48: Shifts in real rate differentials generally explained USD/JPY performance well until the LDP's defeat in the Upper House elections last July



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 49: EM assets sharply decoupled from the Dollar after the spike in 2025

1y correlation of monthly returns with the Dollar Index.

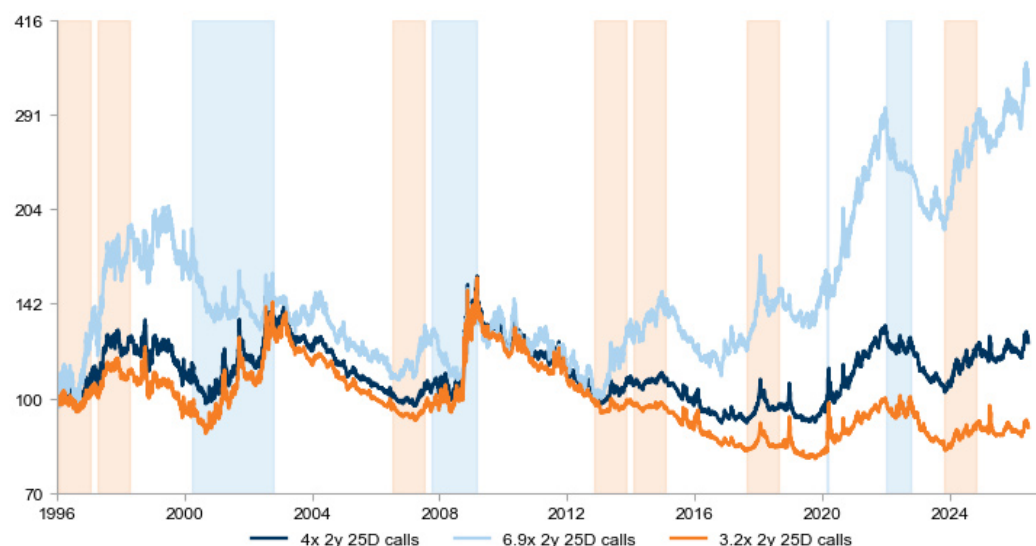


Source: Bloomberg, Datastream, Goldman Sachs Global Investment Research

(4) Keeping all your options — long-dated calls, put spread collars, convertibles
To improve asymmetry or increase upside convexity in a late-cycle boom, we like long-dated call options. As we have shown [before](#), long-dated call options were a very effective tail-risk-management strategy during the Tech Bubble. Replacing equities with a portfolio of 2-year 25-delta call options and IG credit can better tailor risk/reward profiles in late-cycle backdrops. A 4x option position to reach a starting 100-delta generally outperforms during strong rallies and bear markets ([Exhibit 50](#)). Investors can also size the number of options to match the downside tail risk of equities (6.9x) to unlock more upside convexity or create a defensive version (3.2x) to mitigate drawdown risk during bear markets. Compared with short-dated options, longer-dated calls benefit more from sustained equity rallies (particularly when coupled with rising volatility) and carry more efficiently thanks to lower theta decay and a less steep volatility term structure.

Exhibit 50: Long-dated equity calls can improve asymmetry or increase upside convexity

Indexed total return of a portfolio of 2-year calls on S&P 500 (rolled after 1 year) + USD 1-3y IG vs. S&P 500. Orange shade: late-cycle rallies. Blue shade: Bear markets

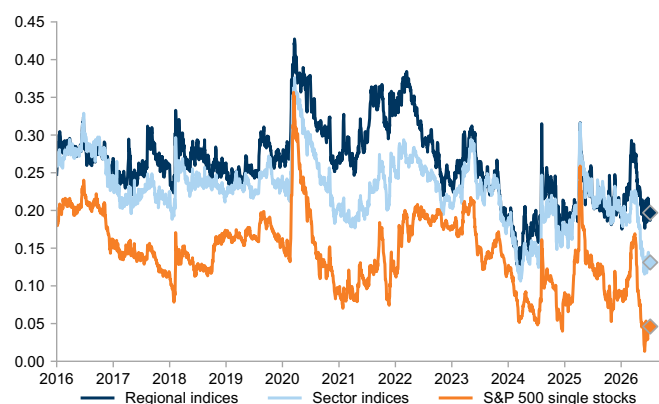


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

In bull markets, investors can also hedge equity exposures with short-dated put spread collars, which can help protect gains buffering smaller corrections. As we showed [before](#), bullish sentiment and strong equity returns are often supported by favourable macro conditions, that can linger. However, the elevated risk of corrections makes put-spreads collars (i.e. selling calls to fund buying put spreads) on equities particularly attractive. This year put/call skew dropped close to historical lows across equity regions, sectors, and single stocks on the back of more bullish option positioning, making collars cheaper ([Exhibit 51](#)). Another smaller asset class that has recently begun to outperform is convertibles, which combine low duration with long equity calls, which can be beneficial in a late-cycle backdrop. Both during the late 1990s and around the GFC, convertibles outperformed vs. a 60/40 portfolio ([Exhibit 52](#)).

Exhibit 51: Put/call skew dropped close to historical lows across equities

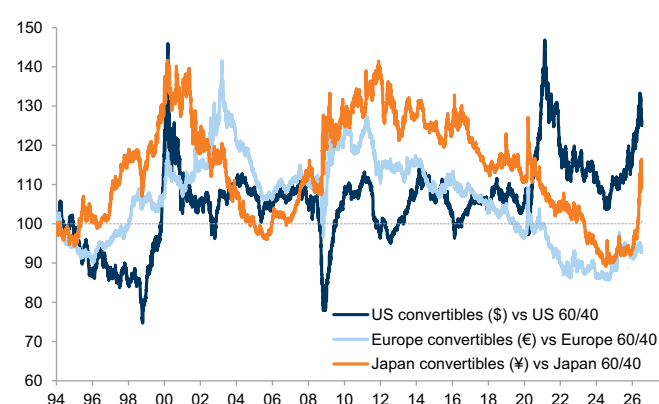
Average 3-month 25-delta put/call skew (normalised)



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 52: Convertibles have started to outperform, which has been typical late cycle

Total return relative performance (rebased to 100)

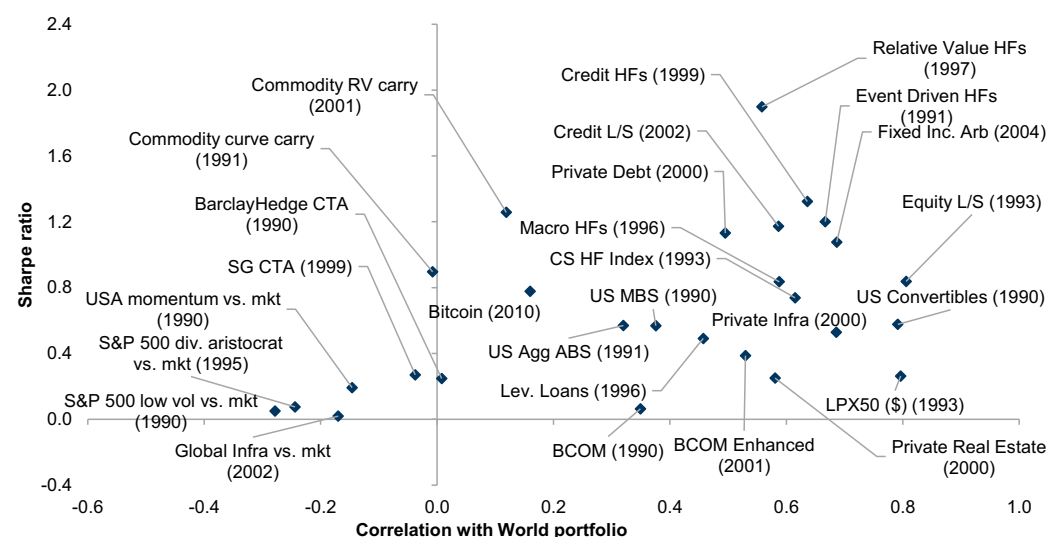


Source: Datastream, Goldman Sachs Global Investment Research

(5) Looking for alternatives — private markets, commodity carry, trend following
Allocations to alternatives can also improve risk-adjusted returns of the World Portfolio in a late-cycle boom. Especially when the return potential and diversification in traditional assets are more limited, alternatives can play a more important role. [Exhibit 53](#) shows several assets and strategies that could enhance the World Portfolio over the medium term through either high Sharpe ratios or diversification benefits. For example equity/long short funds might be able to benefit from rising stock dispersion within the AI ecosystem. Commodity curve carry might also benefit from structurally higher roll yields due to commodity concentration. Finally, private markets may provide access to more targeted and diversified sector and style exposures than public markets benchmarks.

Exhibit 53: Few assets have a low correlation to the World Portfolio but high Sharpe ratio

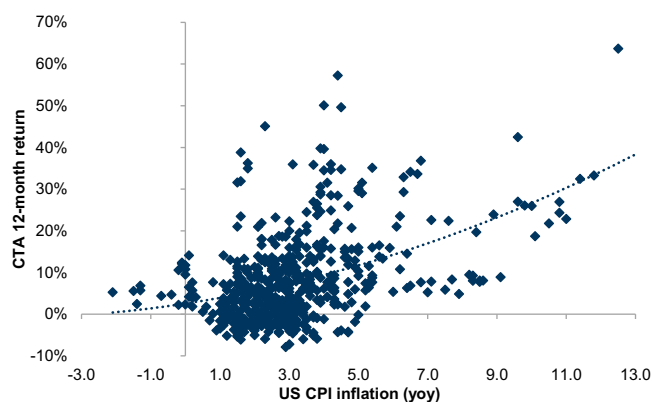
Data since 1990 (where available, start date shown in brackets)



Source: Bloomberg, Datastream, Preqin, Goldman Sachs Global Investment Research

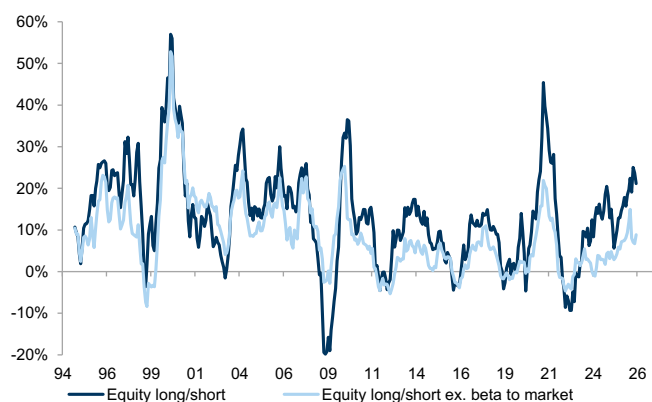
During periods of elevated inflation, CTAs have often delivered stronger returns, irrespective of inflation momentum, in contrast to traditional assets (Exhibit 54). In part, this is due to inflation ‘trending’ more while growth shocks happened fast historically. As a result, investors may underreact initially, and commodities in particular can trend for a long time if supported by fundamentals. Higher inflation also reduces the ability of central banks to buffer the business cycle and drive sharp reversals in markets. In the last cycle, in particular, large dovish pivots by central banks often triggered sharp ‘risk on’ rotations during growth slowdowns. Conversely, ‘risk off’ moves, especially in equities, have been fast with vol of vol trending up post the GFC. Equity long/short hedge funds might help diversify concentrated equity benchmarks and benefit from rising return dispersion linked to AI winners and losers. Historically hedge funds outperformed their beta to the market during late cycle periods and during the Tech Bubble (Exhibit 55).

Exhibit 54: Higher levels of inflation tend to be more supportive of CTA performance



Source: Haver Analytics, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 55: Equity long/short hedge funds did well during the Tech Bubble, also in excess of the market beta
12-month rolling performance and excluding beta to the market



Source: Preqin, Datastream, Goldman Sachs Global Investment Research

References

DeMiguel, V., Garlappi, L. and Uppal, R. 2009. “Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?” *The Review of Financial Studies*, 22 (5): 1915–1953, <https://doi.org/10.1093/rfs/hhm075>

Ielpo, F., Muhammetgulyyeva, S., and Royer, J. 2026. “Building a Regime-Resilient ERC Portfolio.” *The Journal of Portfolio Management*, Early Access.

Kritzman, M., and Turkington, D. 2026. “The Fallacy of Concentration.” *Financial Analysts Journal* 82(2): 8–18.

Michaud, R. 1998. *Efficient Asset Management*. New York: Harvard Business School Press. Now published by Oxford University Press.

Disclosure Appendix

Reg AC

We, Christian Mueller-Glissmann, CFA, Andrea Ferrario and Alessandro Giglio, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Christian Mueller-Glissmann, CFA Goldman Sachs International, Andrea Ferrario Goldman Sachs International, Alessandro Giglio Goldman Sachs International, Giovanni Ferrannini Goldman Sachs International, Elena Porfidia Goldman Sachs International, Peter Oppenheimer Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Option specific disclosures

Price target methodology: Please refer to the analyst's previously published research for methodology and risks associated with equity price targets.

Pricing Disclosure: Option prices and volatility levels in this note are indicative only, and are based on our estimates of recent mid-market levels (unless otherwise noted). All prices and levels exclude transaction costs unless otherwise stated.

General Options Risks: The risks below and any other options risks mentioned in this research report pertain both to specific derivative trade recommendations mentioned and to discussion of general opportunities and advantages of derivative strategies. Unless otherwise noted, options strategies mentioned in this report may be a combination of the strategies below and therefore carry with them the risks of those strategies.

Buying Options: Investors who buy call (put) options risk loss of the entire premium paid if the underlying security finishes below (above) the strike price at expiration. Investors who buy call or put spreads also risk a maximum loss of the premium paid. The maximum gain on a long call or put spread is the difference between the strike prices, less the premium paid.

Selling Options: Investors who sell calls on securities they do not own risk unlimited loss of the security price less the strike price. Investors who sell covered calls (sell calls while owning the underlying security) risk having to deliver the underlying security or pay the difference between the security price and the strike price, depending on whether the option is settled by physical delivery or cash-settled. Investors who sell puts risk loss of the strike price less the premium received for selling the put. Investors who sell put or call spreads risk a maximum loss of the difference between the strikes less the premium received, while their maximum gain is the premium received.

For options settled by physical delivery, the above risks assume the options buyer or seller, buys or sells the resulting securities at the settlement price on expiry.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research

Analyst – SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for “professional investors” within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither “registered banks” nor “deposit takers” (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for “wholesale clients” (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom’s departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International (“GSI”), authorised by the Prudential Regulation Authority (“PRA”) and regulated by the Financial Conduct Authority (“FCA”) and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE (“GSBE”) is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst’s judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or fine-tune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

For more thematic and long-lived research, visit our Themes page >

